

EE104 Fundamentals of Electric Circuits (Lesson 14)

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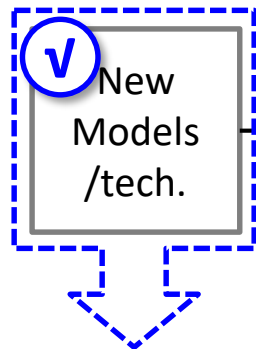
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Review of Lesson 13 (a)

- EE104 road map

Lesson 13



The same laws

The same technique

Lesson 14

The same ways to simplify a circuits (Equivalent circuit)

Transient and full responses
AC circuits

1. Laplace transform
2. Inverse Laplace transform

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
te^{-at}	$\frac{1}{(s+a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Review of Lesson 13 (b)

Type	$f(t) (t > 0-)$	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Functional Transform (函数的拉普拉斯变换)

$$F(x) = \frac{k}{(s + a)^r} \quad f(t) = \frac{kt^{r-1}e^{-at}}{(r-1)!}u(t)$$

$$F(x) = \frac{k}{s + \alpha - j\beta} + \frac{k^*}{s + \alpha + j\beta}$$

$$f(t) = 2|k|e^{-\alpha t} \cos(\beta t + \theta)u(t)$$

$$F(x) = \frac{k}{(s + a - j\beta)^r} + \frac{k^*}{(s + a + j\beta)^r}$$

$$f(t) = \frac{2|k|t^{r-1}e^{-at}}{(r-1)!} \cos(\beta t + \theta)u(t)$$

Review of Lesson 13 (c)

Operation	$f(t)$	$F(s)$
Time integral	$\int_0^t f(x) dx$	$\frac{F(s)}{s}$
Translation in time	$f(t - a)u(t - a), a > 0$	$e^{-as}F(s)$
Translation in frequency	$e^{-at}f(t)$	$F(s + a)$
Scale changing	$f(at), a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
First derivative (s)	$tf(t)$	$-\frac{dF(s)}{ds}$
n th derivative (s)	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s integral	$\frac{f(t)}{t}$	$\int_s^\infty F(u) du$

Operational Transforms
(算子拉普拉斯变换)

Review of Lesson 13 (d)

- Steps to apply inverse Laplace transform:

Possible forms

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}$$

1. When $m > n$, $F(s)$ is a proper rational function (有理真分式)

The roots of $D(s)$
are or include

- Distinct real roots (独立实根)
- Repeated real roots (实重根)
- Distinct complex roots (不等复根)
- Repeat complex roots (多重复根)

2. When $n \leq m$, $F(s)$ is a proper rational function (有理假分式)

Methods

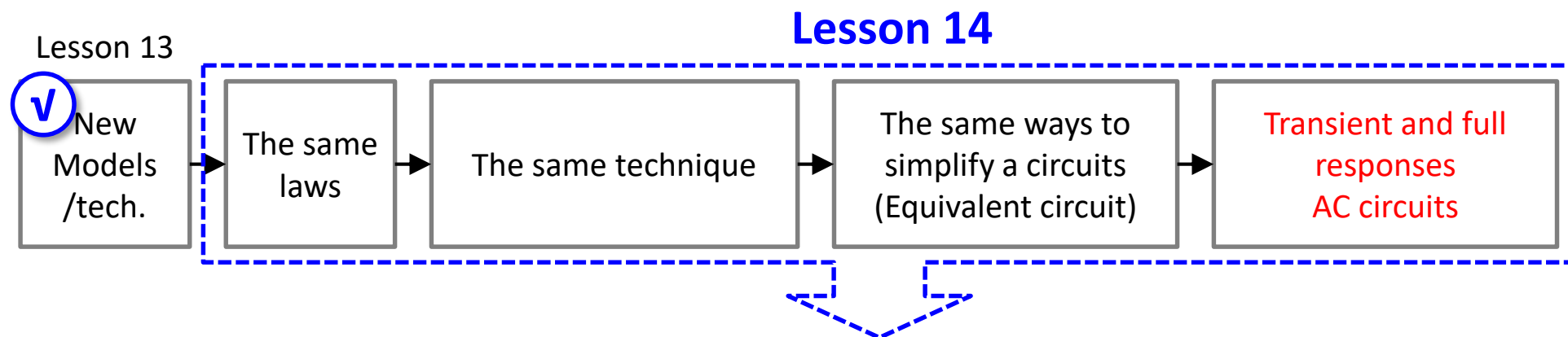
部分分式分解方法

Residue method
(留数法)

Algebraic method
(代数法)

Review of Lesson 14 (e)

- EE104 road map



1. Elements and laws in s domain.
2. Circuits analysis in s domain
3. Transfer function in s domain
4. Transformers in s domain
5. More applications

Lesson 14 Applications of Laplace transform

14.1	Elements and laws in s domain	Slides 08-16
14.2-14.3	Circuit analysis in s domain	Slides 18-46
14.4	Transfer functions in s domain	Slides 48-58
14.5	Transformers in s domain	Slides 60-65
14.6	More applications	Slides 67-73
14.7	Summary and assignment	Slides 75-76

14.1 Elements and laws in s domain (a)

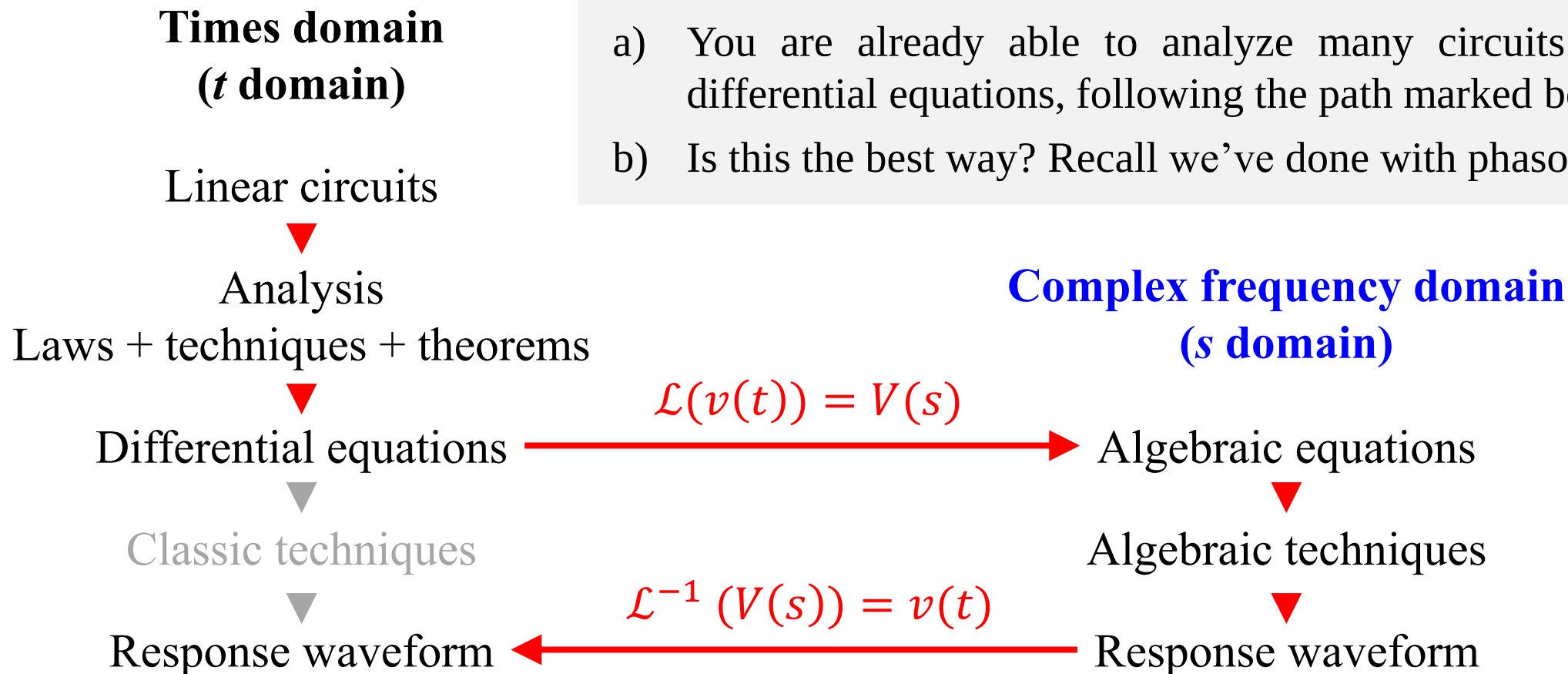
- What we have learned in Lesson 13:

The Laplace transform has two characteristics that make it one of the most powerful mathematical tools for circuits analysis and design.

- 1) it transforms a set of linear constant-coefficient differential equations into a set of ease-handling linear polynomial equations;
- 2) it automatically introduces into the polynomial equations the initial values of the current and voltage variables.

14.1 Elements and laws in s domain (b)

- Laplace transform for circuits analysis: path I

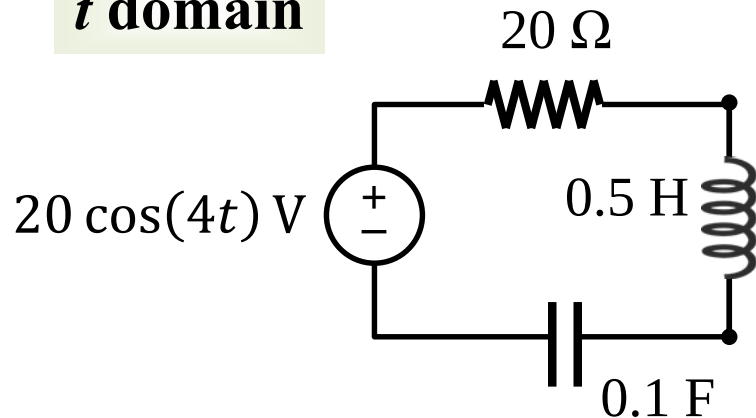


- a) You are already able to analyze many circuits based on differential equations, following the path marked below.
- b) Is this the best way? Recall we've done with phasors.

14.1 Elements and laws in s domain (c)

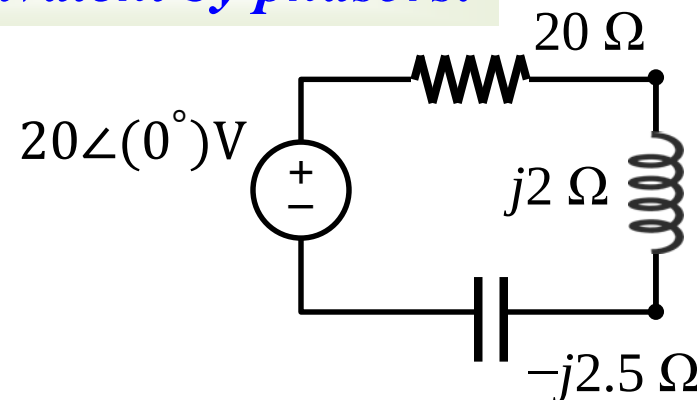
- The equivalent circuit in s domain:

t domain

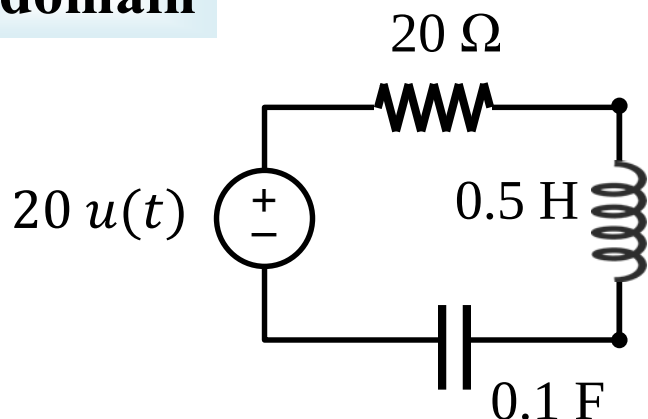


$$\begin{aligned} v(t) &= i(t)R \Rightarrow \mathbf{V} = \mathbf{I}R \\ v(t) &= L di(t)/dt \Rightarrow \mathbf{V} = j\omega L \mathbf{I} \\ v(t) &= \frac{1}{C} \int i(t) dt \Rightarrow \mathbf{V} = \mathbf{I}/j\omega C \end{aligned}$$

Equivalent by phasors:

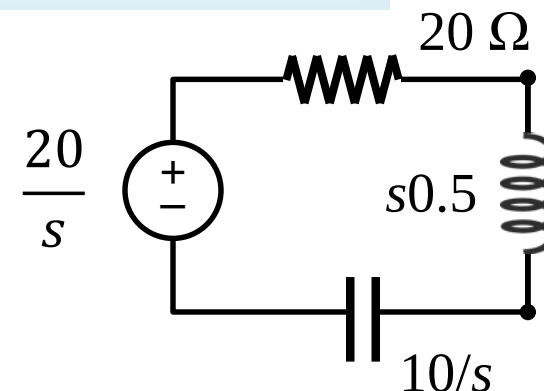


t domain



$$\begin{aligned} v(t) &= i(t)R \Rightarrow \mathbf{V} = \mathbf{I}R \\ v(t) &= L di(t)/dt \Rightarrow \mathbf{V} = sL \mathbf{I} \\ v(t) &= \frac{1}{C} \int i(t) dt \Rightarrow \mathbf{V} = \mathbf{I}/sC \end{aligned}$$

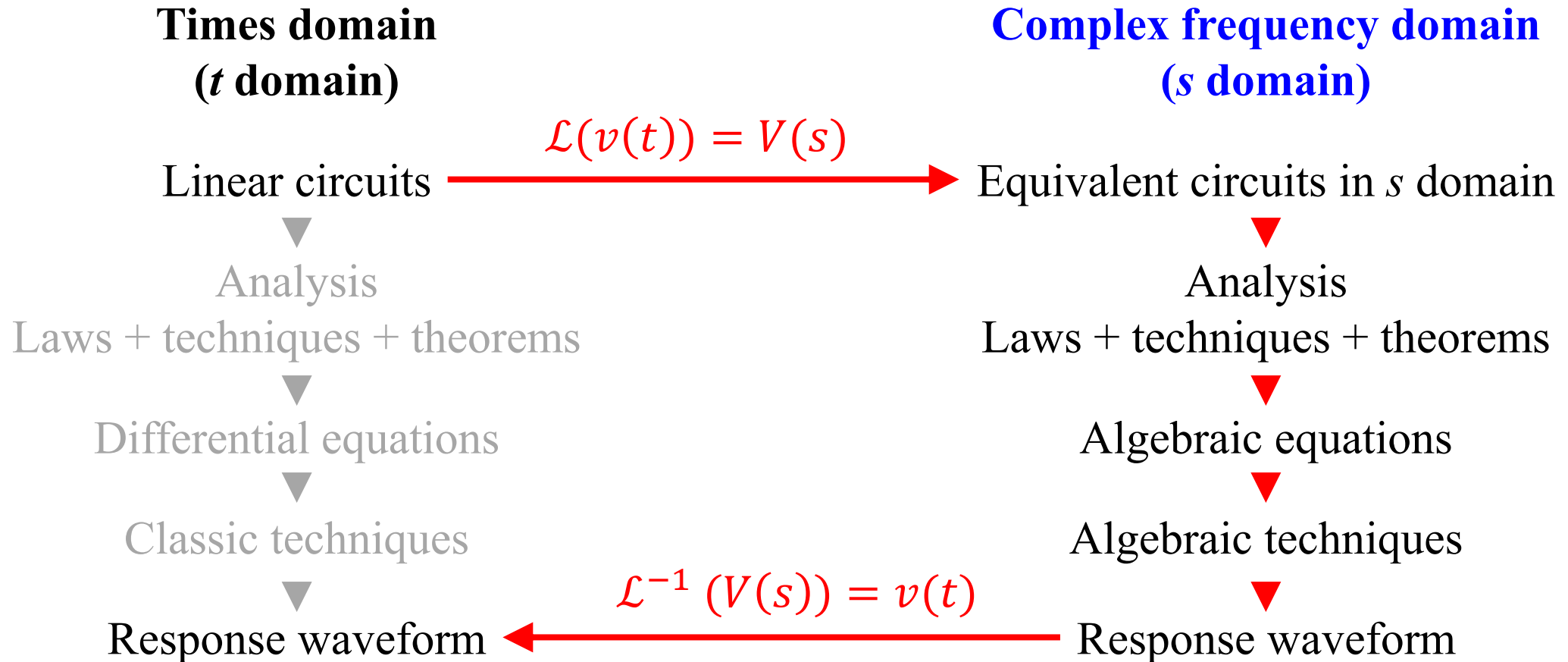
Equivalent in s domain:



*Assuming zero initial conditions

14.1 Elements and laws in s domain (d)

- Laplace transform for circuits analysis: path II



14.1 Elements and laws in s domain (e)

- Resistors are the best!

Time domain

$$v(t) = Ri(t)$$



$$V(s) = RI(s)$$

s domain

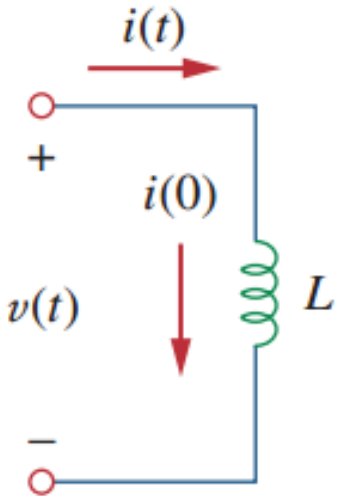


14.1 Elements and laws in s domain (f)

- Inductors in s domain:

Time domain

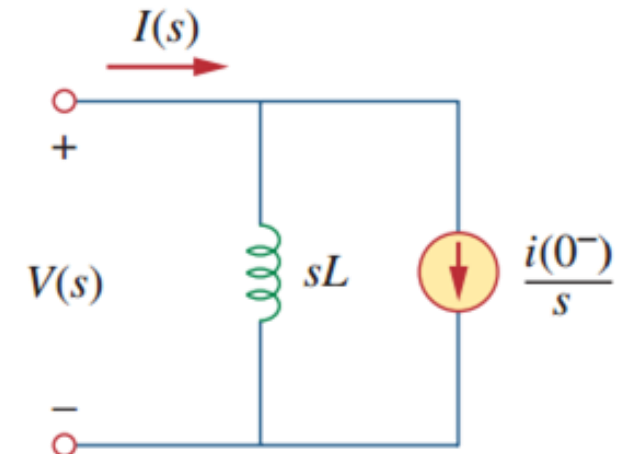
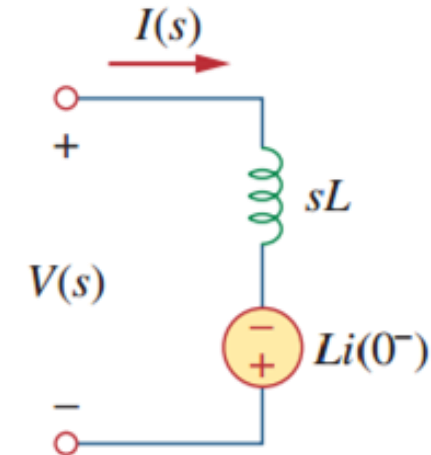
$$v(t) = L \frac{di(t)}{dt}$$



s domain

$$V(s) = L[sI(s) - i(0^-)] \\ = sLI(s) - Li(0^-)$$

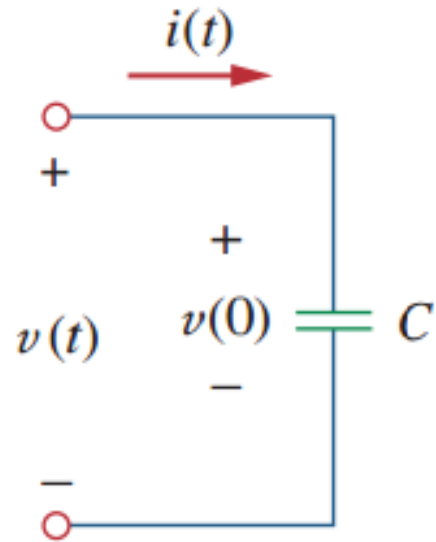
$$I(s) = \frac{1}{sL} V(s) + \frac{i(0^-)}{s}$$



14.1 Elements and laws in s domain (g)

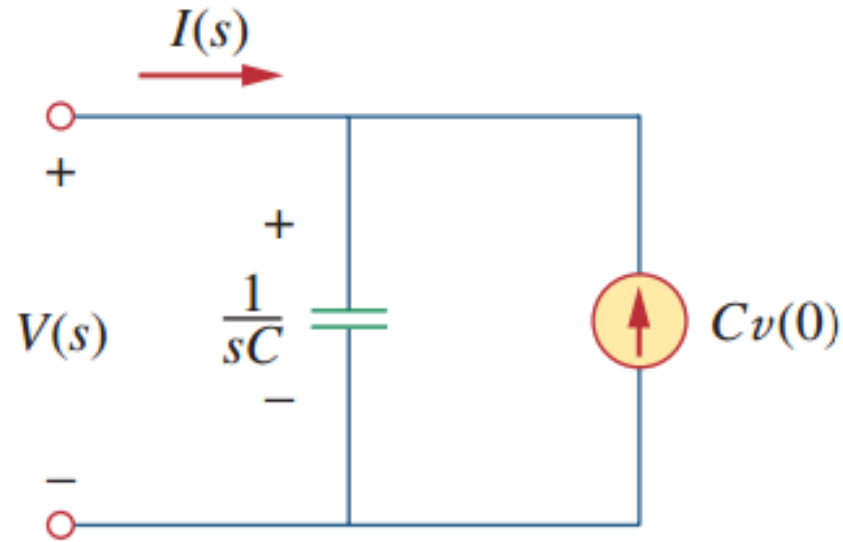
- Capacitors in s domain:

Time domain

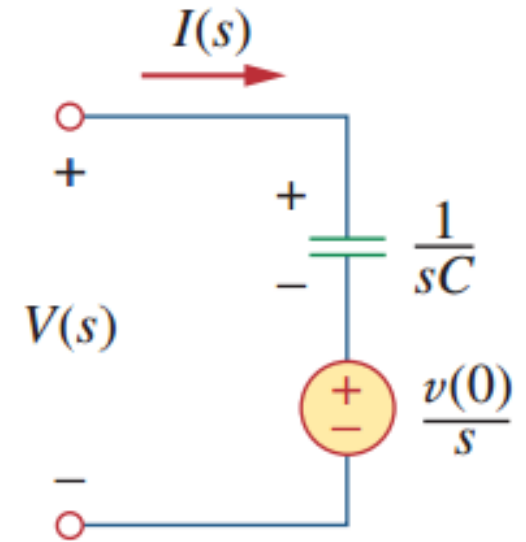


$$i = C \frac{dv}{dt}$$

s domain



$$I = sCV(s) - Cv(0^-)$$



$$V(s) = \frac{1}{sC} I(s) + \frac{v(0^-)}{s}$$

14.1 Elements and laws in s domain (h)

- Ohm's law in s domain:

*Assuming zero initial conditions:

Elements	s domain	f domain	t domain
Resistor	$V = RI$	$\mathbf{V} = \mathbf{R}\mathbf{I}$	$v(t) = i(t)R$
Inductor	$V = sLI$	$\mathbf{V} = j\omega L\mathbf{I}$	$v(t) = L \frac{di(t)}{dt}$
Capacitor	$V = I/sC$	$\mathbf{V} = \mathbf{I}/j\omega C$	$v(t) = \int Ci(t) dt$

Impedance in s domain:

$$Z(s) = \frac{V(s)}{I(s)} \begin{cases} \text{Resistor:} & Z(s) = R \\ \text{Inductor:} & Z(s) = sL \\ \text{Capacitor:} & Z(s) = \frac{1}{sC} \end{cases}$$

Admittance in s domain:

$$Y(s) = \frac{1}{Z(s)} = \frac{I(s)}{V(s)}$$

14.1 Elements and laws in s domain (i)

- Dependent sources in s domain :

- b) The models for dependent sources are easy to develop drawing from the simple fact that if the Laplace transform of $f(t)$ is $F(s)$, then the Laplace transform of $af(t)$ is $aF(s)$ —the linearity property.

$$\mathcal{L}[av(t)] = aV(s) \quad \mathcal{L}[ai(t)] = aI(s)$$

- KCL and KVL in s domain :

- a) All the circuit theorems and relationships developed for DC circuits are perfectly valid in the s-domain. The s-domain version of Kirchhoff's laws is:

$$\Sigma I(s) = 0; \quad \Sigma V(s) = 0$$

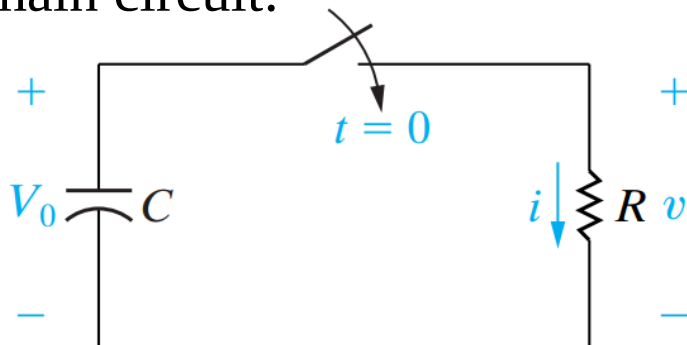
Lesson 14 Applications of Laplace transform

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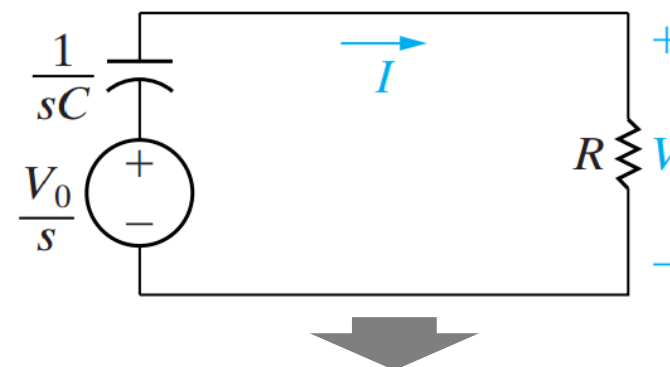
14.2 Circuit analysis in s domain (a)

- How to analyze a circuit in s domain: RC circuits

A t -domain circuit:



Step 1: an s -domain equivalent circuit.



Step 3: inverse Laplace transform:

$$i = \frac{V_0}{R} e^{-t/RC} u(t),$$
$$v = Ri = V_0 e^{-t/RC} u(t).$$

The natural response of an RC circuit

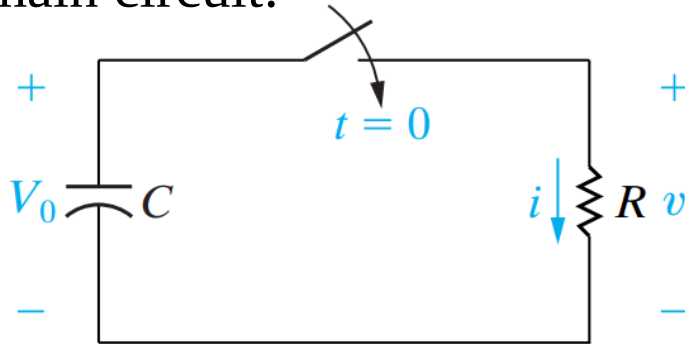
Step 2: using KVL and solve the equation

$$\frac{V_0}{s} = \frac{1}{sC} I + RI.$$
$$I = \frac{CV_0}{RCs + 1} = \frac{V_0/R}{s + (1/RC)}$$

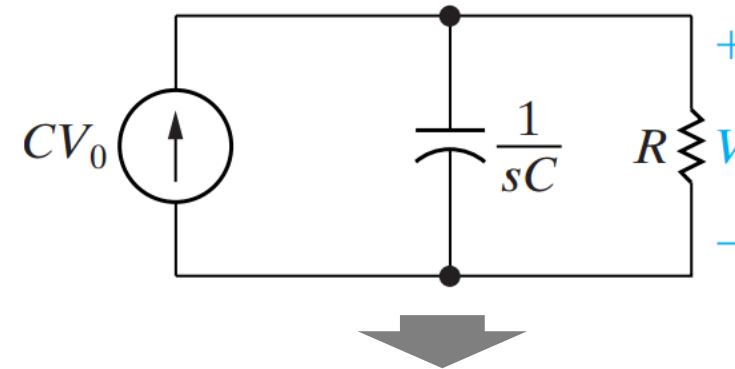
14.2 Circuit analysis in s domain (b)

- How to analyze a circuit in s domain: RC circuits

A t -domain circuit:



Step 1: an s-domain equivalent circuit.



Step 3: inverse Laplace transform:

$$i = \frac{V_0}{R} e^{-t/RC} u(t),$$
$$v = Ri = V_0 e^{-t/RC} u(t).$$

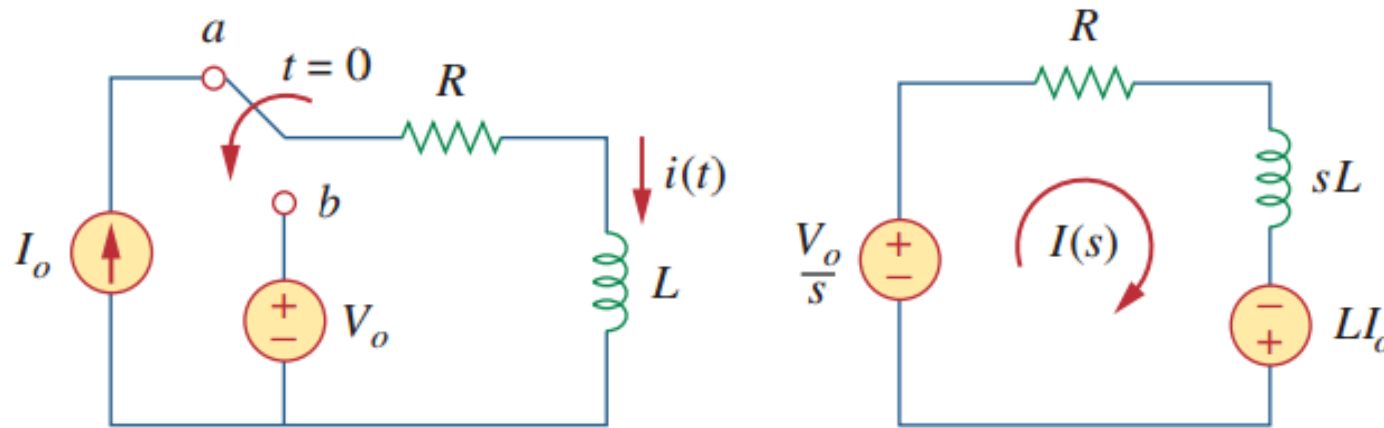
The natural response of an RC circuit

Step 2: using KCL and solve the equation

$$\frac{V}{R} + sCV = CV_0.$$
$$V = \frac{V_0}{s + (1/RC)}.$$

14.2 Circuit analysis in s domain (c)

- RL circuits:



The initial current through the inductor is $i(0) = I_o$

$$I(s)(R + sL) - LI_o - \frac{V_o}{s} = 0 \quad \Rightarrow \quad I(s) = \frac{I_o}{s + R/L} + \frac{V_o/R}{s} - \frac{V_o/R}{(s + R/L)}$$

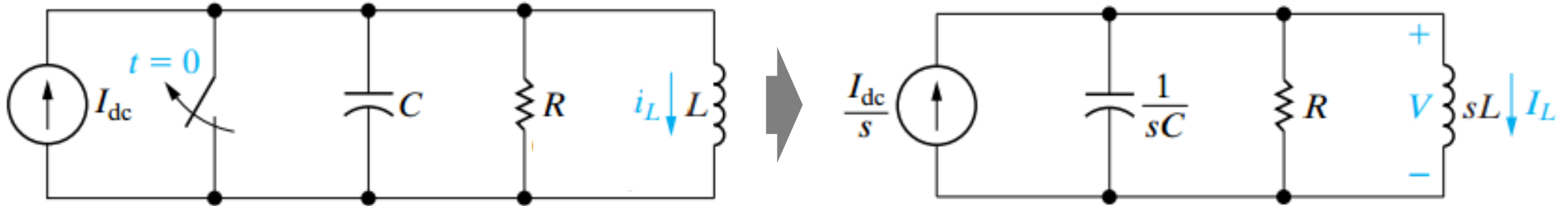
$$i(t) = \left(I_o - \frac{V_o}{R} \right) e^{-t/\tau} + \frac{V_o}{R}, \quad t \geq 0$$

Transient response

Steady-state response

14.2 Circuit analysis in s domain (d)

- RLC circuits:



Initial values are zero, sing KCL, we have: $sCV + \frac{V}{R} + \frac{V}{sL} = \frac{I_{dc}}{s}$

Solving for V gives: $V = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$

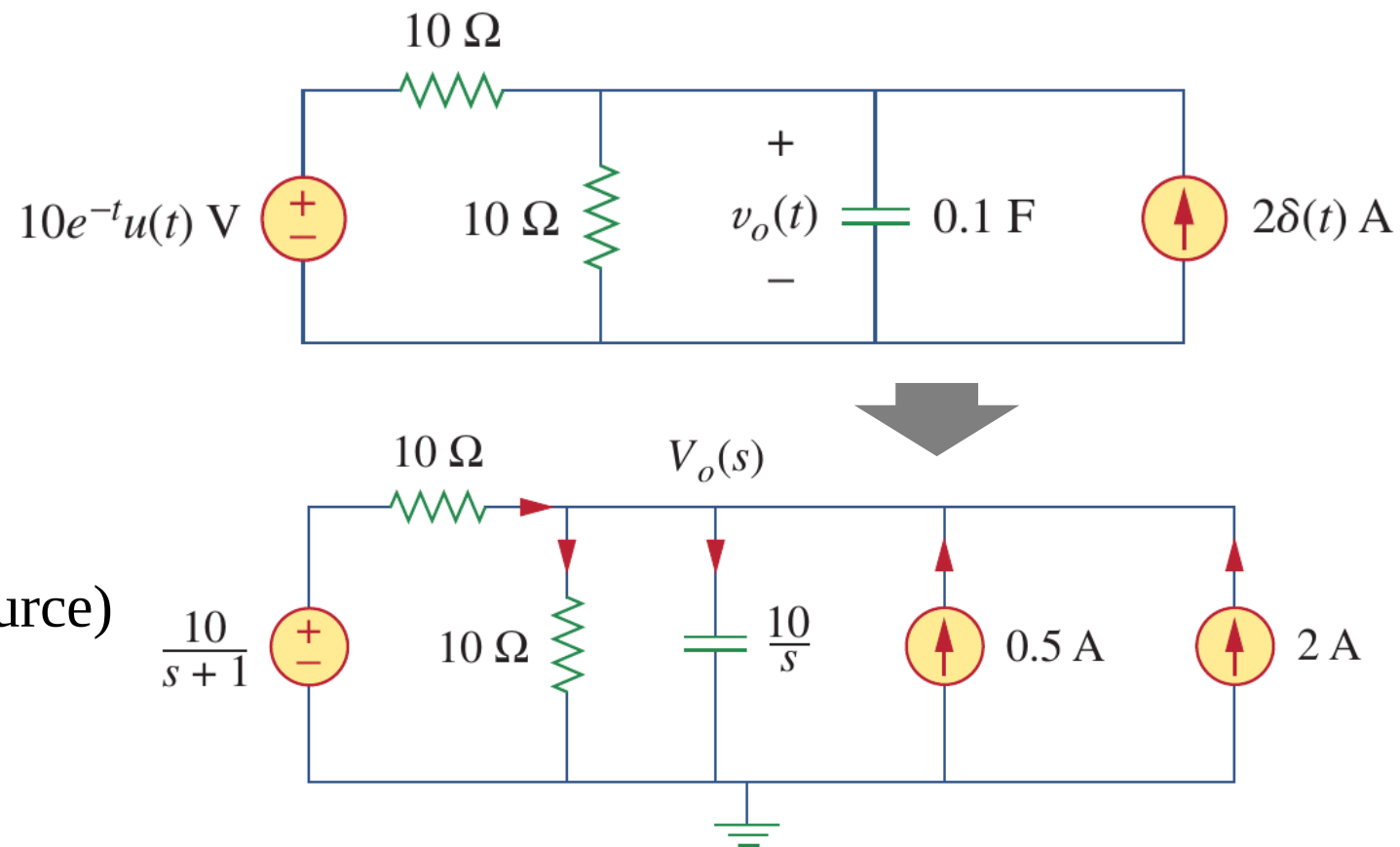
Thus the inductor current: $I_L = \frac{I_{dc}/LC}{s[s^2 + (1/RC)s + (1/LC)]}$

14.2 Circuit analysis in s domain (e)

- Example 14.01:

Example 14.01: find $v_o(t)$ in the circuit, assuming $v_o(0) = 5$ V.

Solution: (1) find the equivalent circuit in s domain:



$$\left\{ \begin{array}{ll} 10e^{-t}u(t) \Rightarrow \frac{10}{s+1} & \text{(the voltage source)} \\ 0.1\text{ F} \Rightarrow \frac{1}{sC} = \frac{10}{s} & \text{(the capacitor)} \\ 2\delta(t) \Rightarrow 2\text{ A} & \text{(the current source)} \end{array} \right.$$

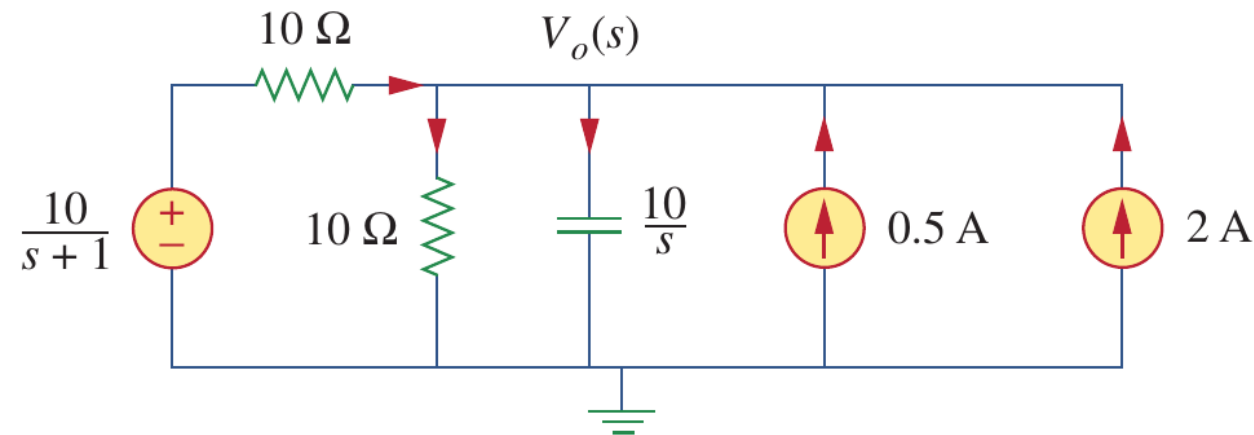
DON'T forget the initial value: $I_c(0) = Cv_o(0) = 0.5$ A

14.2 Circuit analysis in s domain (f)

- **Example 14.01:**

Example 14.01: find $v_o(t)$ in the circuit, assuming $v_o(0) = 5$ V.

Solution: (2) solve for $V_o(s)$:



$$\frac{10/(s+1) - V_o}{10} + 2 + 0.5 = \frac{V_o}{10} + \frac{V_o}{10/s} \Rightarrow V_o = \frac{25s + 35}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$\begin{cases} A = (s+1)V_o(s) \big|_{s=-1} = \frac{25s+35}{(s+2)} \big|_{s=-1} = \frac{10}{1} = 10 \\ B = (s+2)V_o(s) \big|_{s=-2} = \frac{25s+35}{(s+1)} \big|_{s=-2} = \frac{-15}{-1} = 15 \end{cases} \Rightarrow V_o(s) = \frac{10}{s+1} + \frac{15}{s+2}$$

14.2 Circuit analysis in s domain (g)

- Example 14.01:

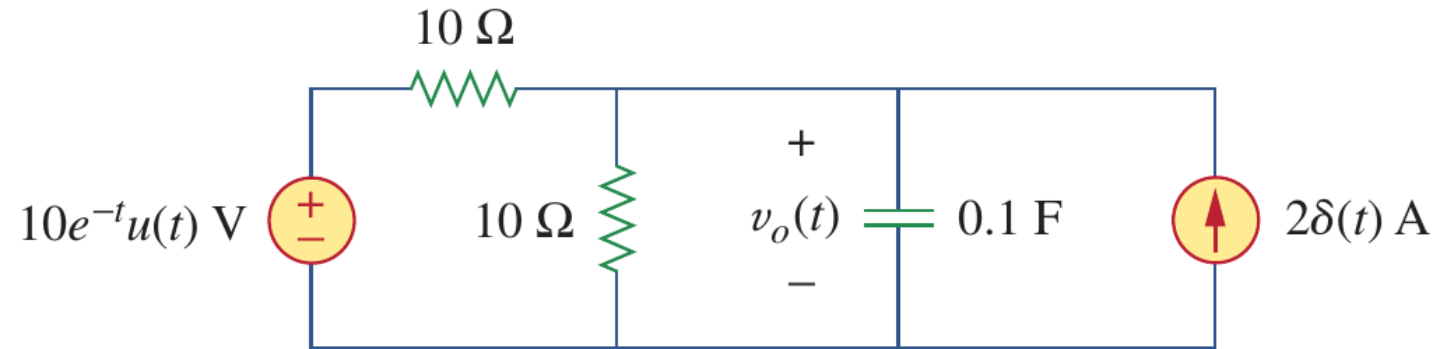
Example 14.01: find $v_o(t)$ in the circuit, assuming $v_o(0) = 5$ V.

Solution: (3) get back into the time:

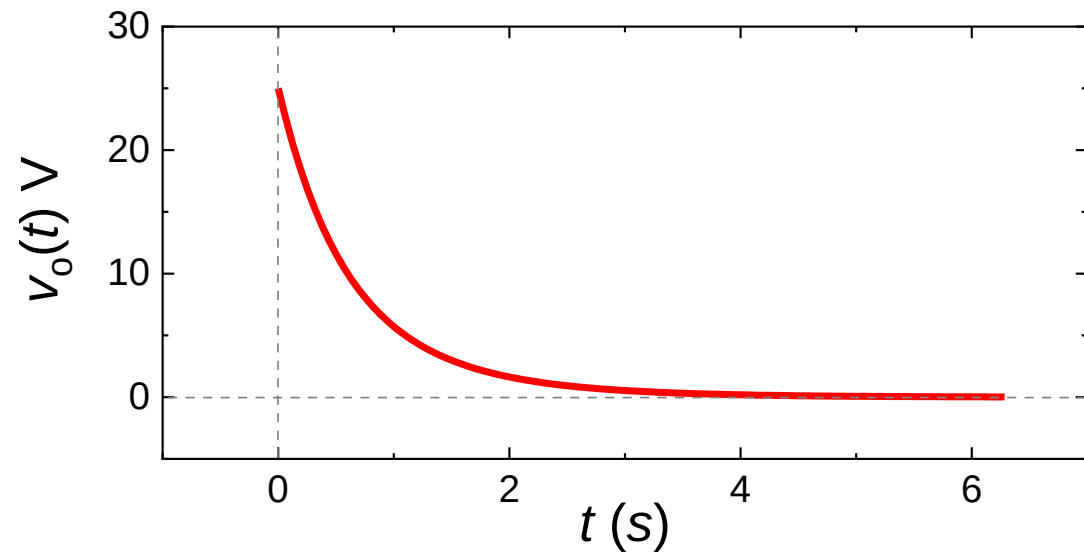
$$V_o(s) = \frac{10}{s+1} + \frac{15}{s+2}$$



$$v_o(t) = (10e^{-t} + 15e^{-2t})u(t) \text{ V}$$



(4) Here you can see the waveform:



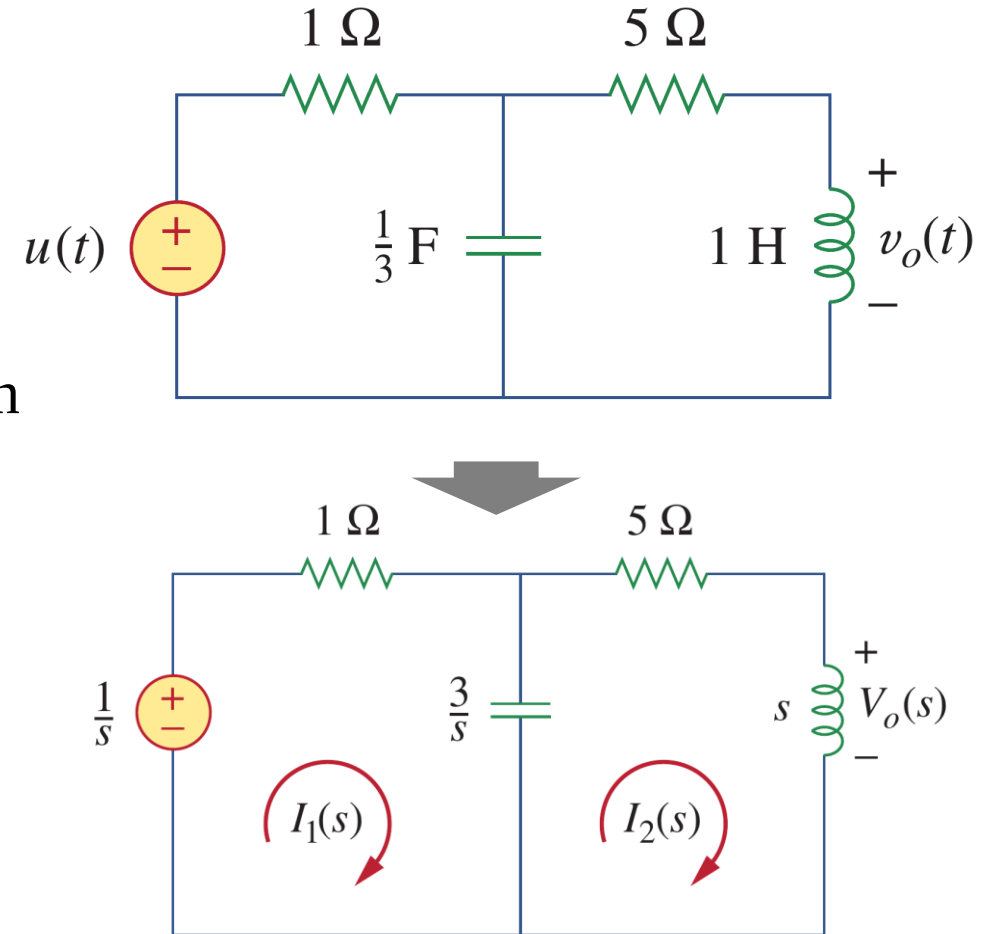
14.2 Circuit analysis in s domain (h)

- Example 14.02:

Example 14.02: find $v_o(t)$ in the following circuit, assuming zero initial conditions.

Solution: (1) transform the circuit from the time domain to the s domain. Note initial values are zero here.

$$\begin{aligned} u(t) &\Rightarrow \frac{1}{s} \\ 1 \text{ H} &\Rightarrow sL = s \\ \frac{1}{3} \text{ F} &\Rightarrow \frac{1}{sC} = \frac{3}{s} \end{aligned}$$



14.2 Circuit analysis in s domain (i)

- **Example 14.02:**

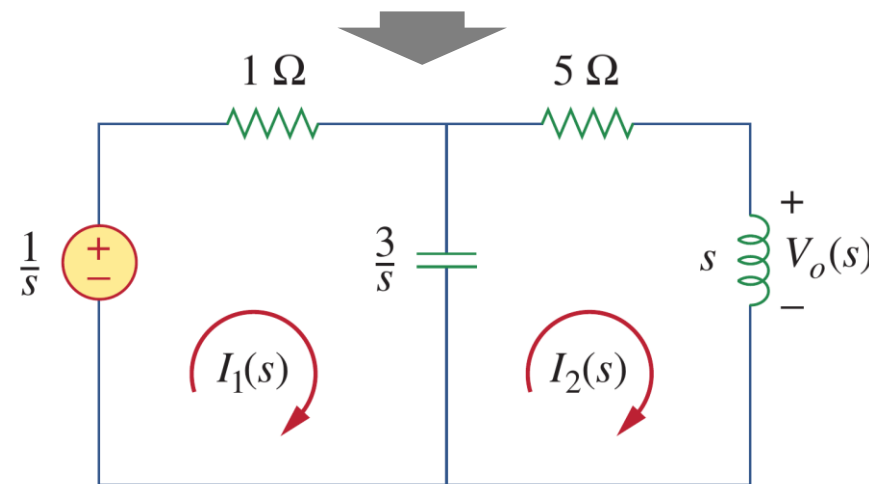
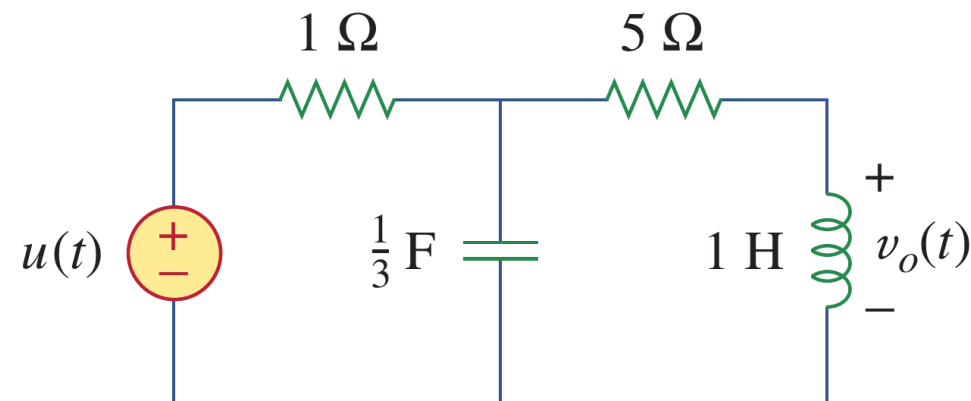
Example 14.02: find $v_o(t)$ in the following circuit, assuming zero initial conditions.

Solution: (2) find for $V_o(s)$ using mesh analysis:

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right)I_1 - \frac{3}{s}I_2 \quad 0 = -\frac{3}{s}I_1 + \left(s + 5 + \frac{3}{s}\right)I_2$$

$$\Rightarrow I_2 = \frac{3}{s^3 + 8s^2 + 18s}$$

$$V_o(s) = sI_2 = \frac{3}{s^2 + 8s + 18} = \boxed{\frac{3}{\sqrt{2}} \frac{\sqrt{2}}{(s + 4)^2 + (\sqrt{2})^2}}$$



Another way of applying inverse Laplace transform in addition to the residue approach.

14.2 Circuit analysis in s domain (j)

• Example 14.02:

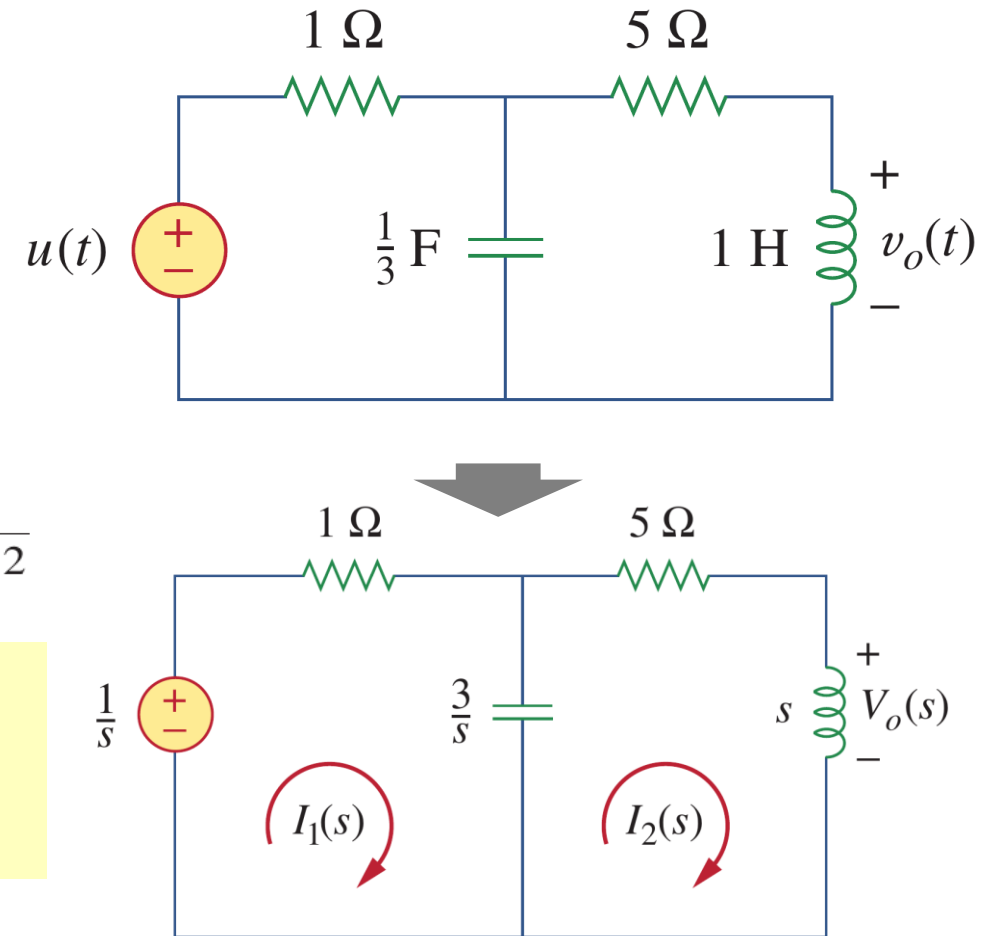
Example 14.02: find $v_o(t)$ in the following circuit, assuming zero initial conditions.

Solution: (3) get $v_o(t)$ using inverse Laplace transform:

$$V_o(s) = sI_2 = \frac{3}{s^2 + 8s + 18} = \frac{3}{\sqrt{2}} \frac{\sqrt{2}}{(s + 4)^2 + (\sqrt{2})^2}$$

$$\begin{cases} \text{Recall the transform pair: } \mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2} \\ \text{Recall the frequency shift: } \mathcal{L}(e^{-at} f(t)) = F(s + a) \end{cases}$$

$$v_o(t) = \frac{3}{\sqrt{2}} e^{-4t} \sin(\sqrt{2}t) \text{ V}, \quad t \geq 0$$

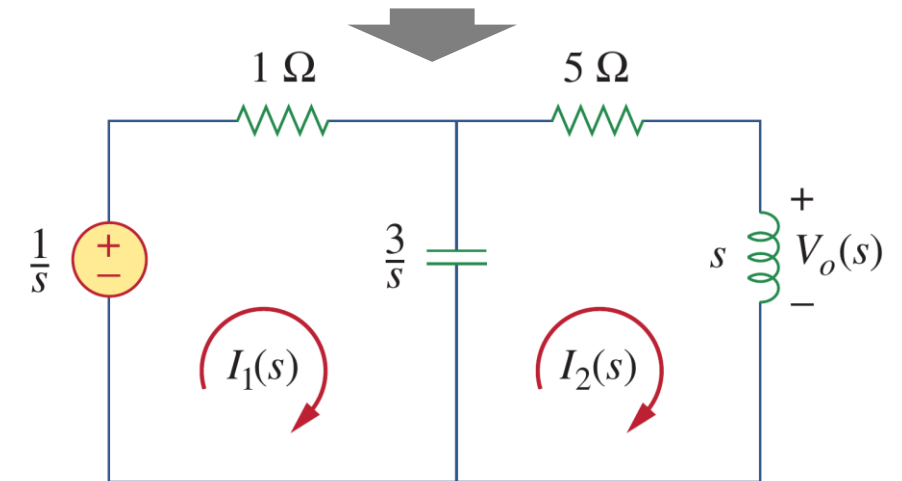
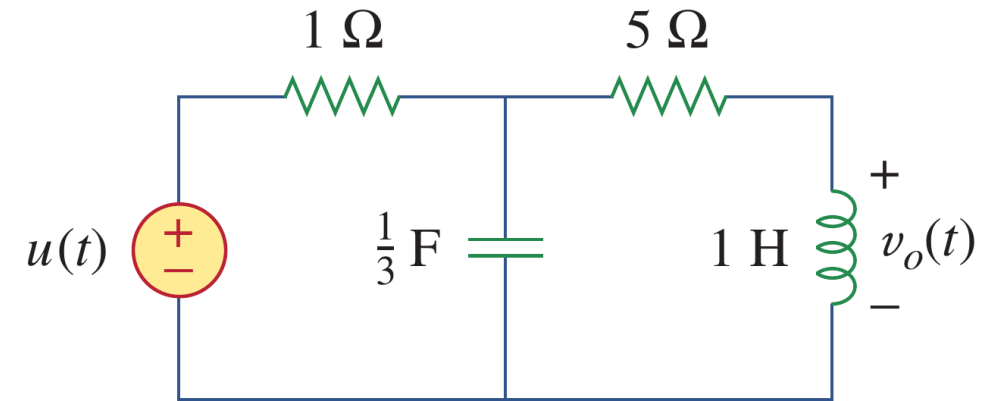
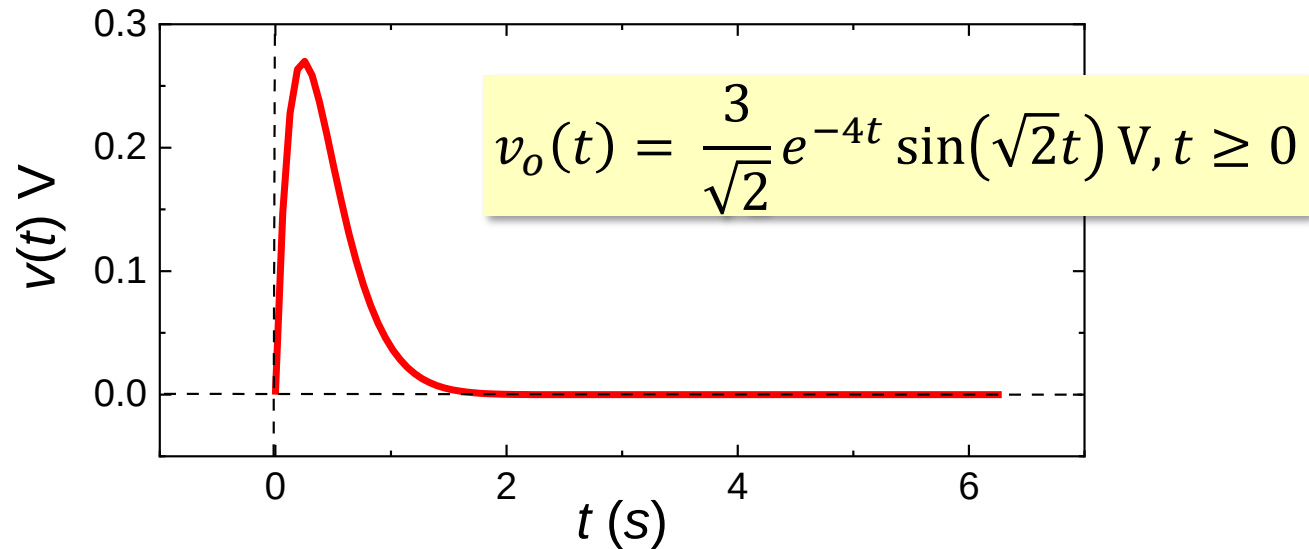


14.2 Circuit analysis in s domain (k)

- Example 14.02:

Example 14.02: find $v_o(t)$ in the following circuit, assuming zero initial conditions.

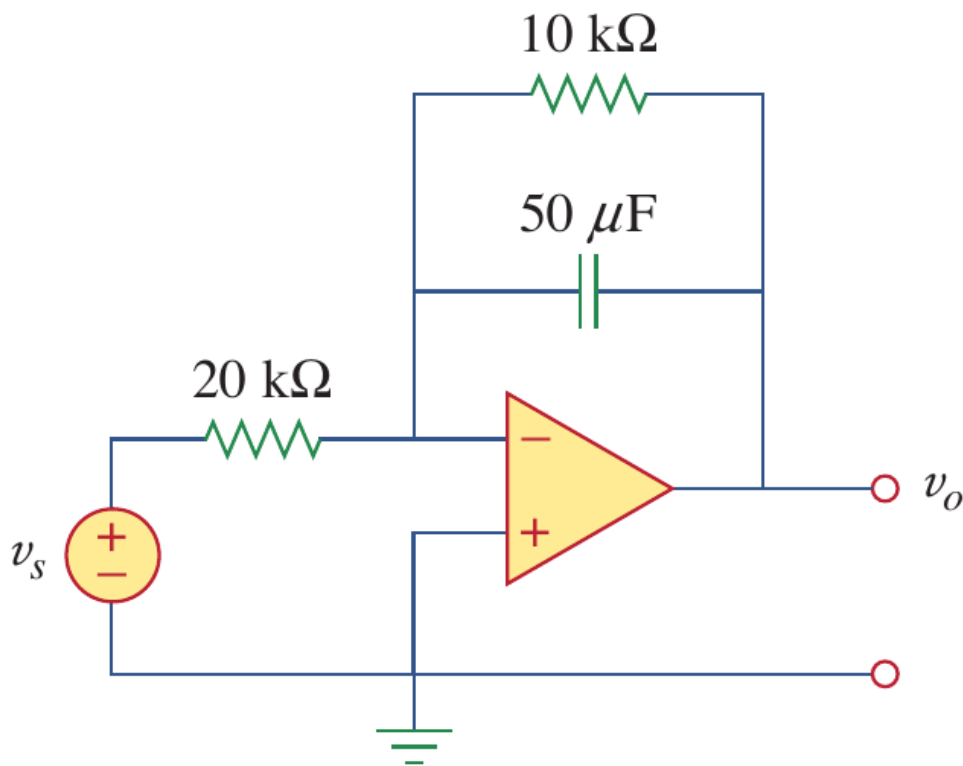
Solution: (4) here you can see the waveform:



14.2 Circuit analysis in s domain (I)

- Example 14.03:

Example 14.03: find $v_o(t)$ for $t > 0$ for the following op amp circuit. $v_s = 3e^{-5t}u(t)$ V.



Solution: (1) find the equivalent in the s domain and use nodal analysis as well as the golden rules:

$$V_s = \frac{3}{s+5}, \quad R_1 = 20\text{ k}\Omega, R_2 = 10\text{ k}\Omega, C = 50\text{ }\mu\text{F}$$

$$\frac{V_s - 0}{R_1} = \frac{0 - V_o}{R_2} + \frac{0 - V_o}{\frac{1}{sC}} \Rightarrow V_s = R_1 \left(\frac{1}{R_2} + sC \right) (-V_o)$$
$$\Rightarrow \frac{V_o}{V_s} = (-1) / \left(sR_1C + \frac{R_1}{R_2} \right)$$

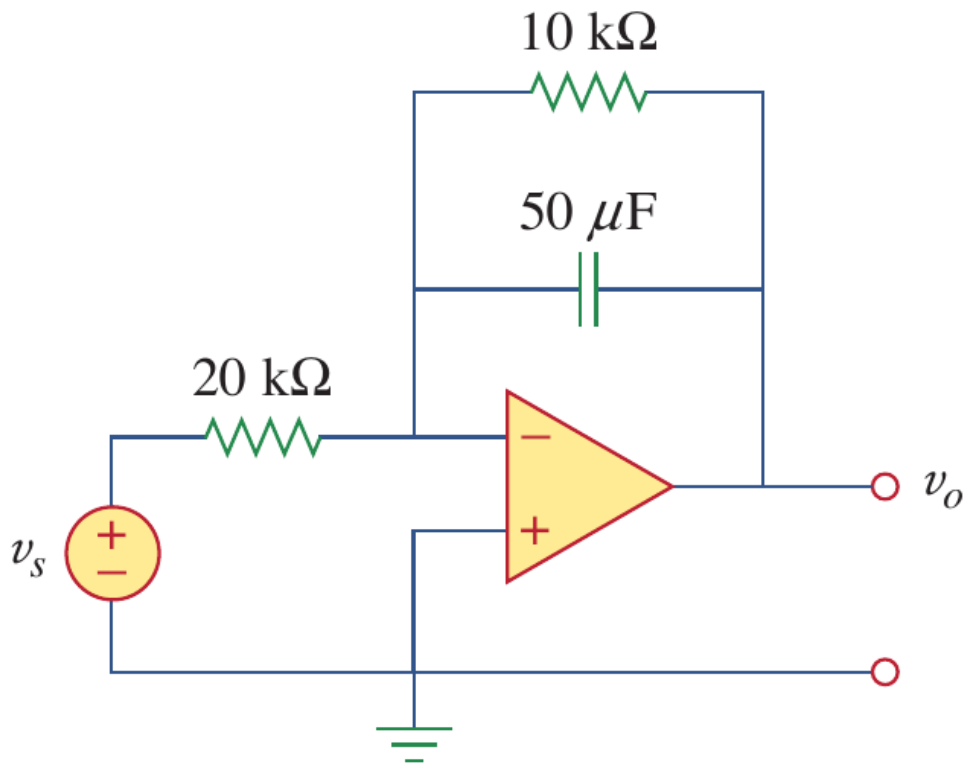
$$R_1/R_2 = 2, R_1C = 1 \Rightarrow \frac{V_o}{V_s} = \frac{-1}{s+2} = \text{gain} = H(s)$$

In s domain

14.2 Circuit analysis in s domain (m)

- Example 14.03:

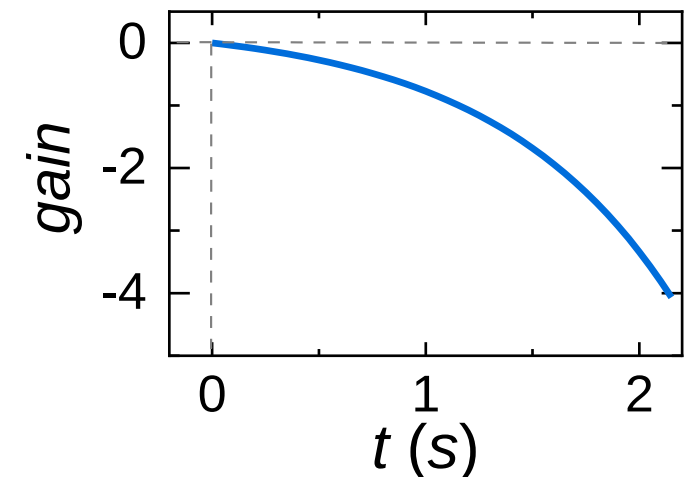
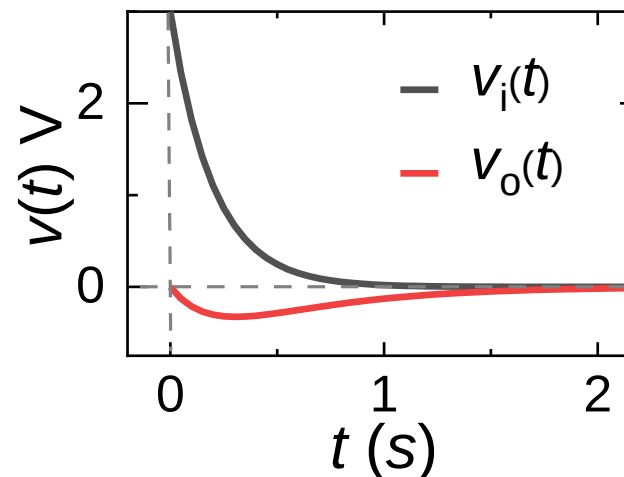
Example 14.03: find $v_o(t)$ for $t > 0$ for the following op amp circuit. $v_s = 3e^{-5t}u(t)$ V.



Solution: (2) find $V_o(s)$ and $v_o(t)$:

$$\frac{V_o}{V_s} = \frac{-1}{s+2} \Rightarrow V_o = \frac{-1}{s+2} \frac{3}{s+5} = \frac{1}{s+5} - \frac{1}{s+2}$$

$$\Rightarrow v_o(t) = \mathcal{L}^{-1}(V_o) = [e^{-5t} - e^{-2t}]u(t) \text{ V}$$

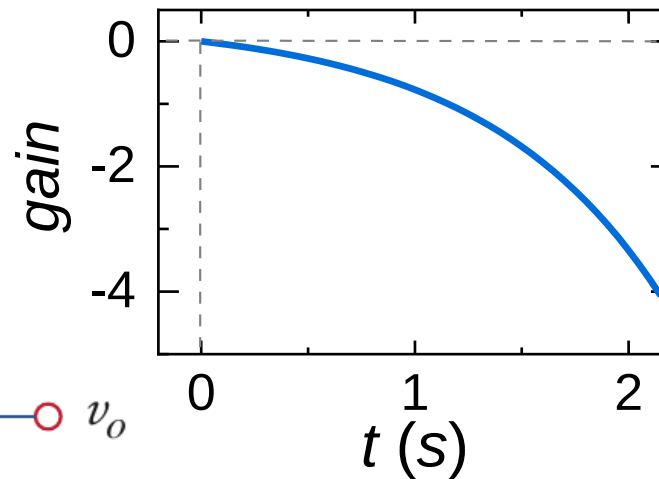
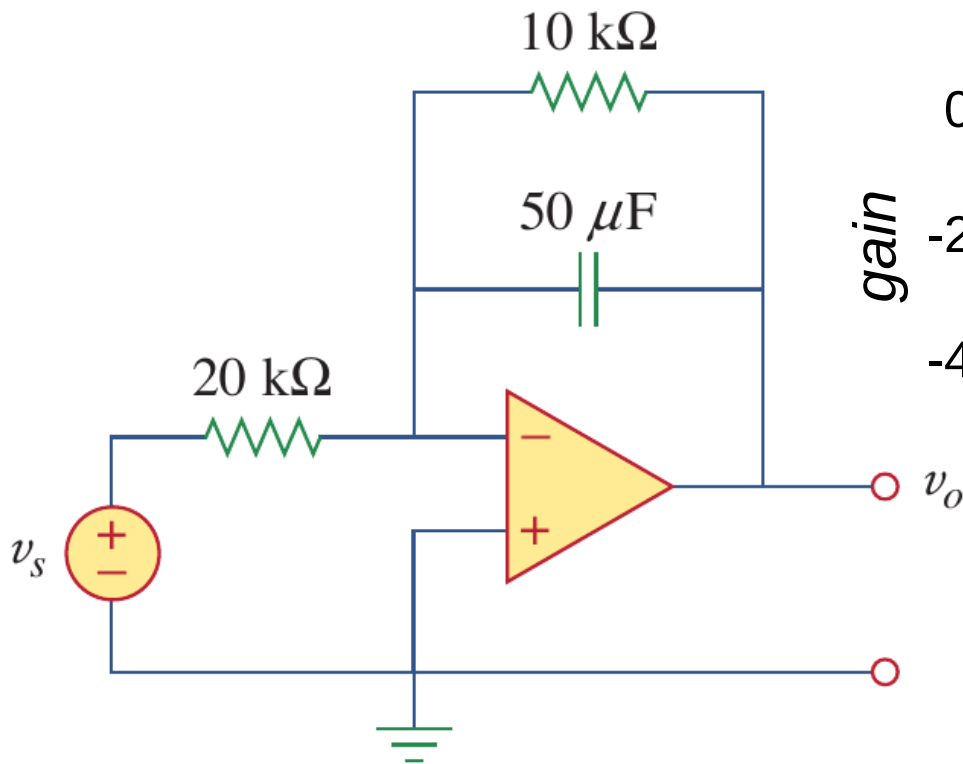


14.2 Circuit analysis in s domain (n)

- Example 14.03:

Example 14.03: find $v_o(t)$ for $t > 0$ for the following op amp circuit. $v_s = 3e^{-5t}u(t)$ V.

Solution: (3) Note that the gain varies with time:



$$\frac{v_o(t)}{v_s(t)} = \frac{e^{-5t} - e^{-2t}}{3e^{-5t}} = \frac{1}{3}(1 - e^{3t})$$

$$\mathcal{L}^{-1}\left(\frac{V_o}{V_s}\right) = \mathcal{L}^{-1}\left(\frac{-1}{s+2}\right) = -e^{-2t}u(t)$$

The inverse Laplace transform of the gain in s domain is not the gain in time domain.
What is it? We will see later.

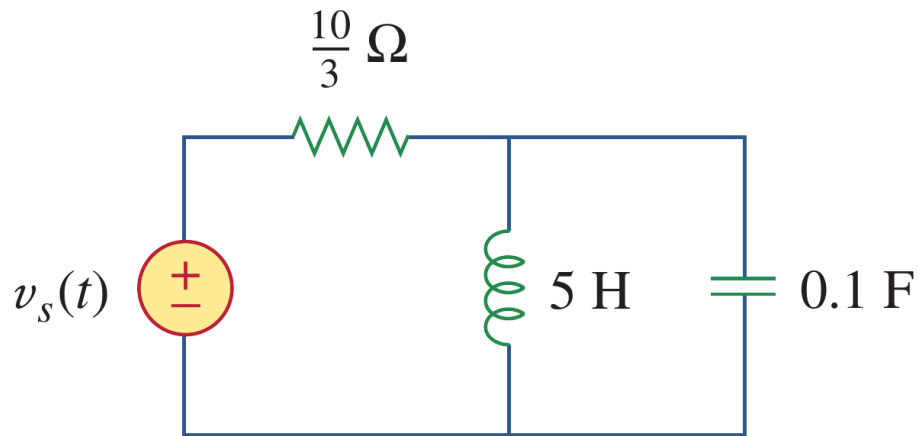
14.3 Circuit theorems in s domain (a)

- Circuit theorems in s domain:

- a) Circuit analysis is again relatively easy to do when we are in the s -domain.
- b) All of the circuit theorems and relationships we developed for dc circuits are perfectly valid in the s -domain.

14.3 Circuit theorems in s domain (b)

- Example 14.03 (superposition theorem):



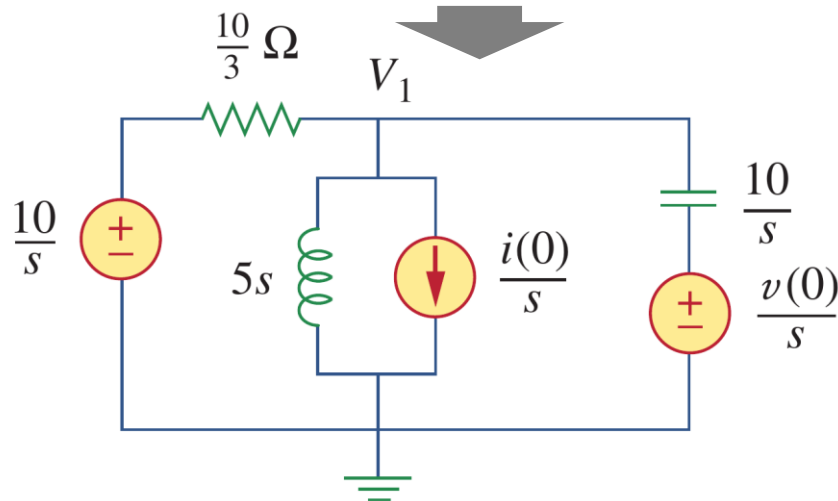
Example 14.03: Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10u(t)$ V. Assume that at $t = 0$, -1 A flows through the inductor and $+5$ V is across the capacitor.

Solution I: (1) Find the equivalent in s domain:

$$\begin{cases} v_s(t) = 10u(t) \Rightarrow V(s) = 10/s \\ 5 \text{ H} \Rightarrow 5s \\ 0.1 \text{ F} \Rightarrow 10/s \end{cases}$$

DON'T forget the initial value:

$$I_L(0) = i_L(0)/s = -1/s \text{ \& } V_c(0) = v_c(0)/s = 5/s$$

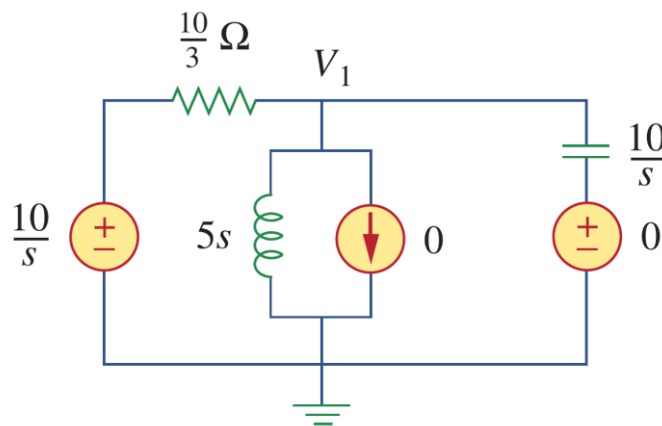


14.3 Circuit theorems in s domain (c)

- Example 14.03 (superposition theorem):

Solution I: (2) Use superposition theorem to solve for $v_1(t)$ & $v_2(t)$ & $v_3(t)$:

Only the voltage source:

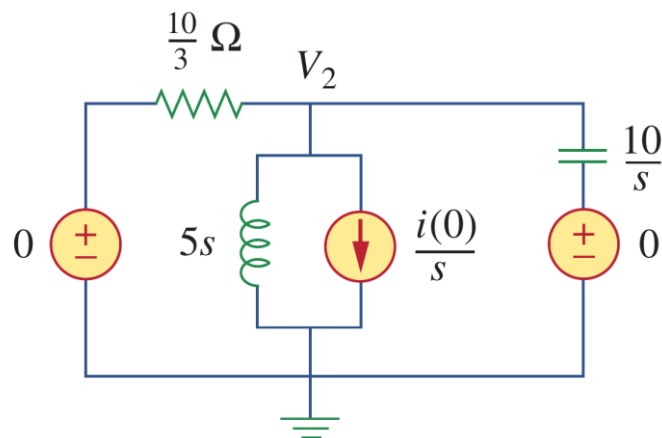


$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + 0 + \frac{V_1 - 0}{1/(0.1s)} = 0$$

$$V_1 = \frac{30}{(s+1)(s+2)} = \frac{30}{s+1} - \frac{30}{s+2}$$

$$v_1(t) = (30e^{-t} - 30e^{-2t})u(t) \text{ V}$$

Initial inductor current:



$$\frac{V_2 - 0}{10/3} + \frac{V_2 - 0}{5s} - \frac{1}{s} + \frac{V_2 - 0}{1/(0.1s)} = 0$$

$$V_2 = \frac{10}{(s+1)(s+2)} = \frac{10}{s+1} - \frac{10}{s+2}$$

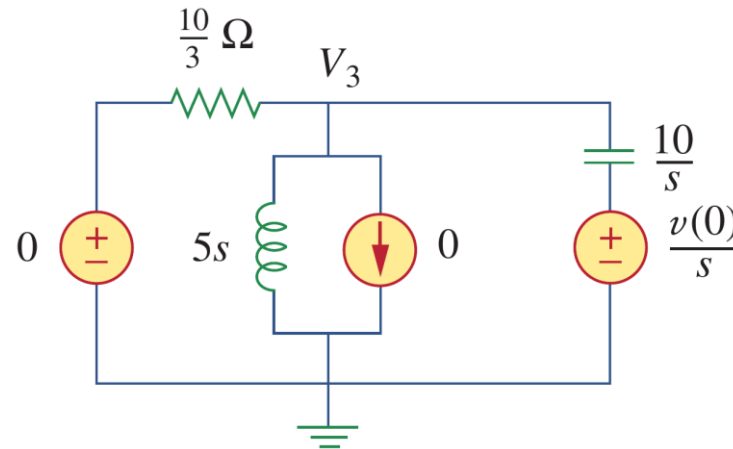
$$v_2(t) = (10e^{-t} - 10e^{-2t})u(t) \text{ V}$$

14.3 Circuit theorems in s domain (d)

- **Example 14.03 (superposition theorem):**

Solution I: (2) Use superposition theorem to solve for $v_1(t)$ & $v_2(t)$ & $v_3(t)$:

Initial capacitor voltage:



$$\frac{V_3 - 0}{10/3} + \frac{V_3 - 0}{5s} - 0 + \frac{V_3 - 5/s}{1/(0.1s)} = 0$$

$$V_3 = \frac{5s}{(s+1)(s+2)} = \frac{-5}{s+1} + \frac{10}{s+2}$$

$$v_3(t) = (-5e^{-t} + 10e^{-2t})u(t) \text{ V}$$

(3) Sum up $v_1(t)$, $v_2(t)$, and $v_3(t)$ to find $v_c(t)$:



$$v(t) = v_1(t) + v_2(t) + v_3(t)$$

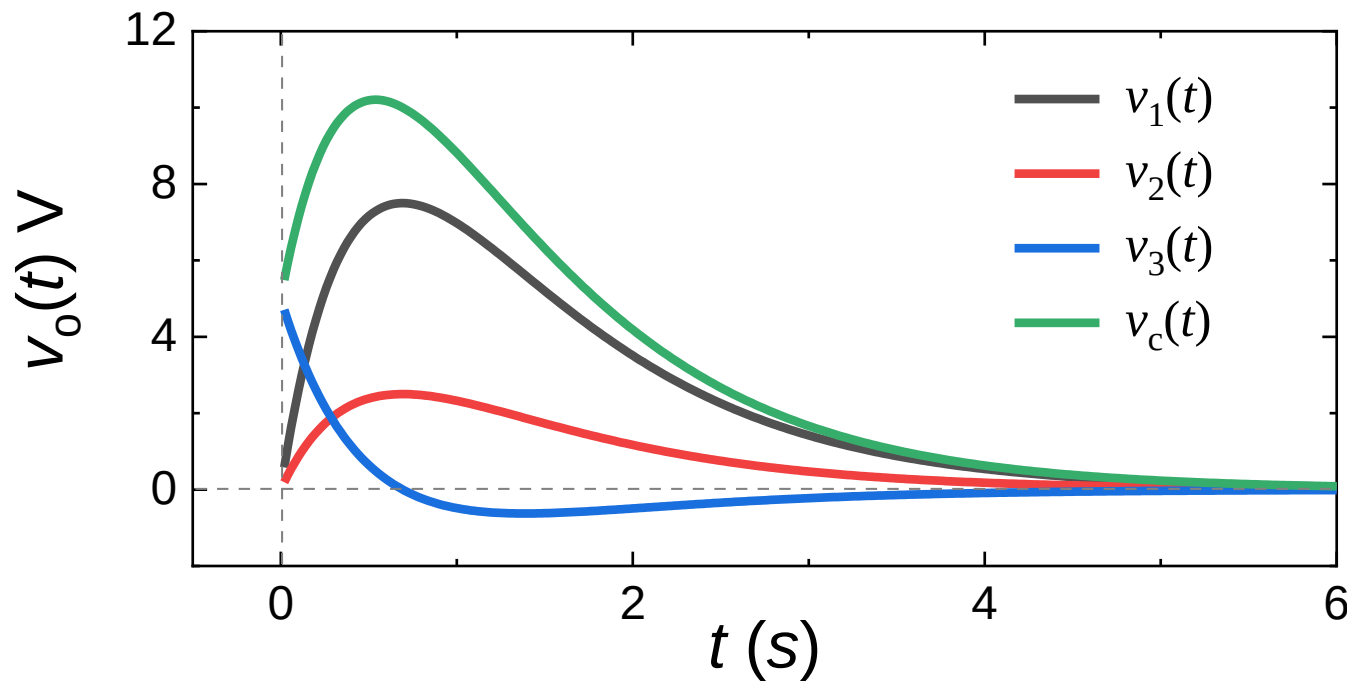
$$= \{(30 + 10 - 5)e^{-t} + (-30 + 10 - 10)e^{-2t}\}u(t) \text{ V}$$

$$v(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V}$$

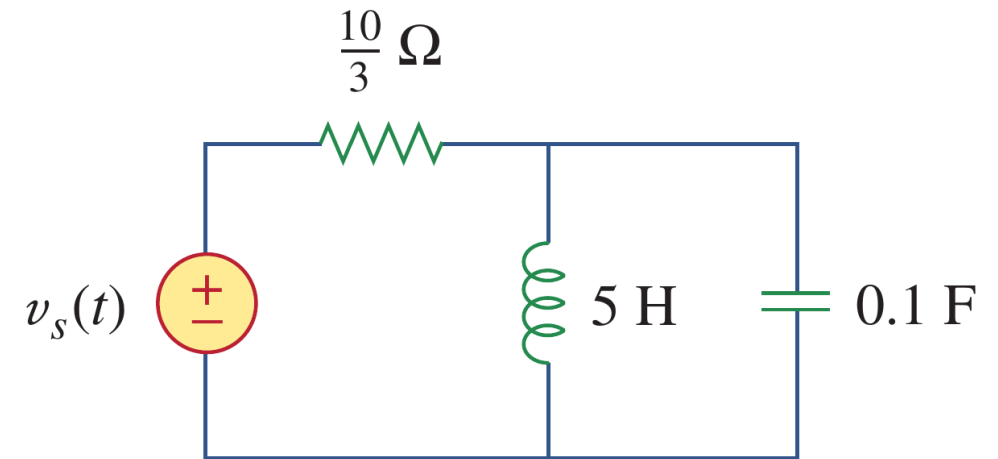
14.3 Circuit theorems in s domain (e)

- Example 14.03 (superposition theorem):

(4) The waveforms are like this:

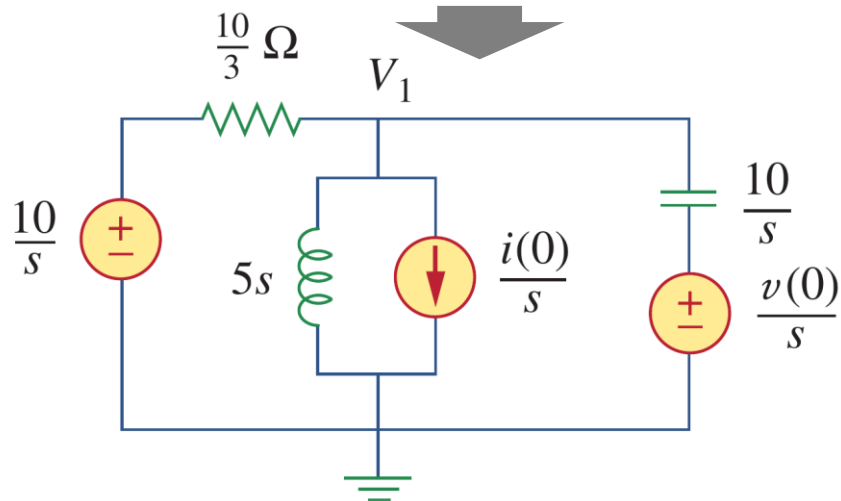
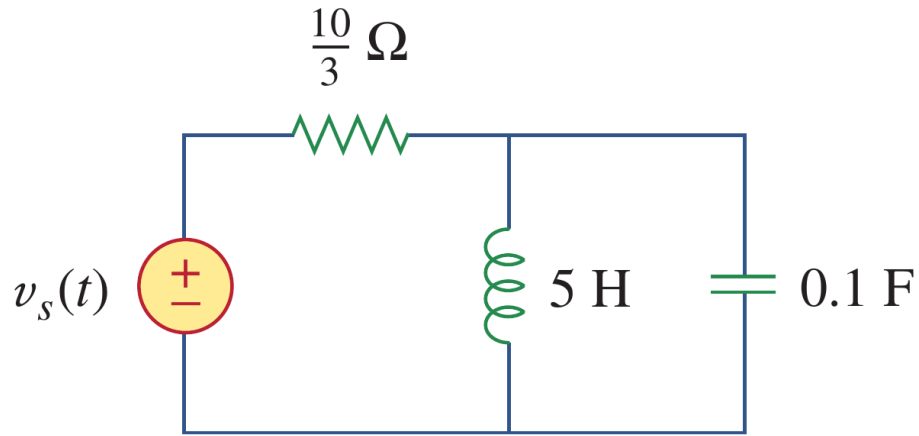


$$v(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V}$$



14.3 Circuit theorems in s domain (f)

- Example 14.03:



Example 14.03: Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10u(t)$ V. Assume that at $t = 0$, -1 A flows through the inductor and $+5$ V is across the capacitor.

Solution II: you can also use nodal analysis

$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + \frac{i(0)}{s} + \frac{V_1 - [v(0)/s]}{1/(0.1s)} = 0$$

$$V_1 = \frac{40 + 5s}{(s + 1)(s + 2)} = \frac{35}{s + 1} - \frac{30}{s + 2}$$

$$v_1(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V}$$

14.3 Circuit theorems in s domain (g)

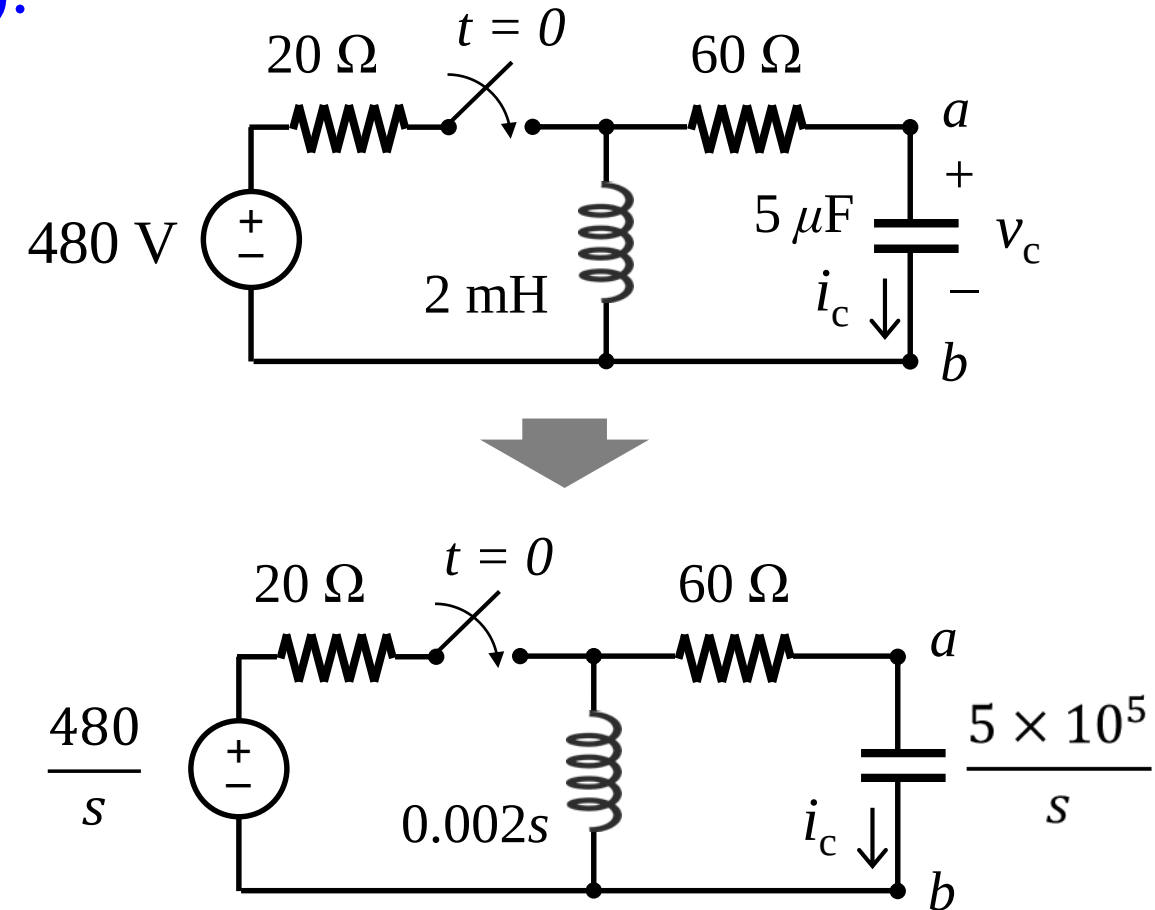
- **Example 14.04 (Thevenin theorem):**

Example 14.04: Find $i_c(t)$ in the following circuit, assuming the energy stored in the circuit prior to the closing is zero.

Solution: (1) get the equivalent in the s domain:

$$\begin{cases} 480 \text{ V at } t = 0 \Rightarrow V(s) = 480/s \\ 2 \text{ mH} \Rightarrow 0.002s \\ 0.1 \text{ F} \Rightarrow 5 \times 10^5/s \end{cases}$$

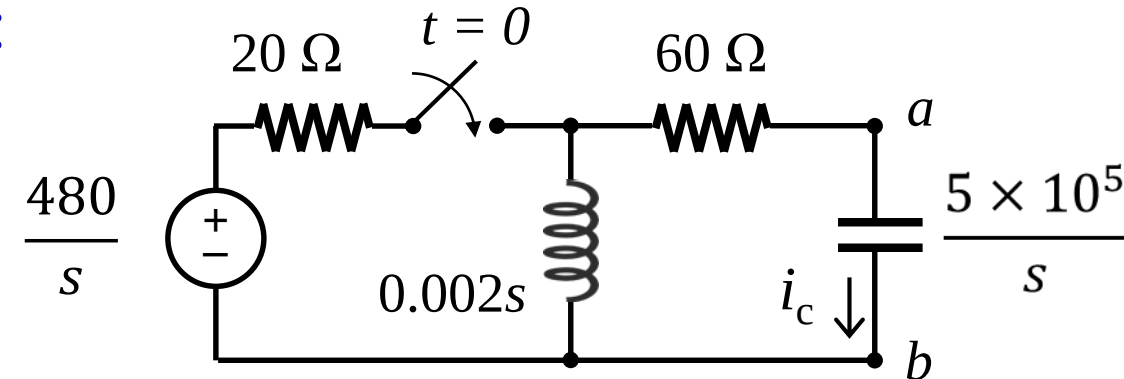
DON'T forget the initial values, which are “zero here as the energy stored in the circuit prior to the closing is zero”.



14.3 Circuit theorems in s domain (h)

- **Example 14.04 (Thevenin theorem):**

Example 14.04: Find $i_c(t)$ in the following circuit, assuming the energy stored in the circuit prior to the closing is zero.



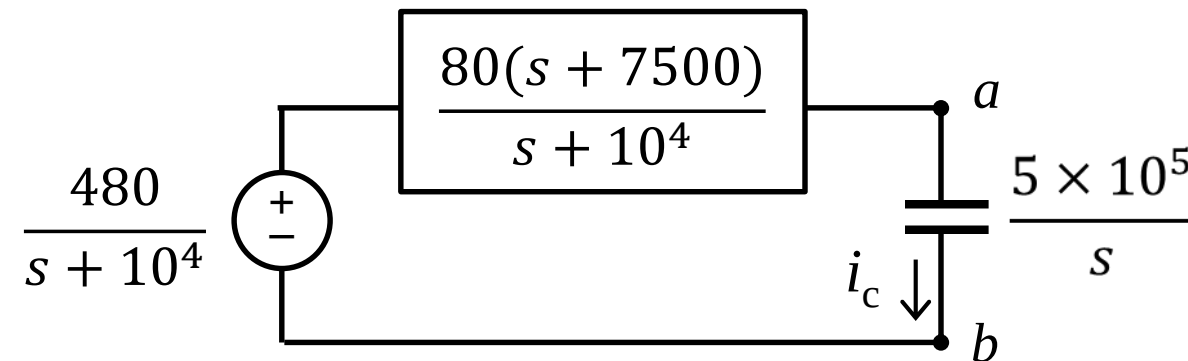
Solution: (2) find the Thevenin equivalent in the s domain:

The Thevenin voltage:

$$V_{Th} = \frac{(480/s)(0.002s)}{20 + 0.002s} = \frac{480}{s + 10^4}$$

The Thevenin impedance:

$$Z_{Th} = 60 + \frac{0.002s(20)}{20 + 0.002s} = \frac{80(s + 7500)}{s + 10^4}$$



14.3 Circuit theorems in s domain (i)

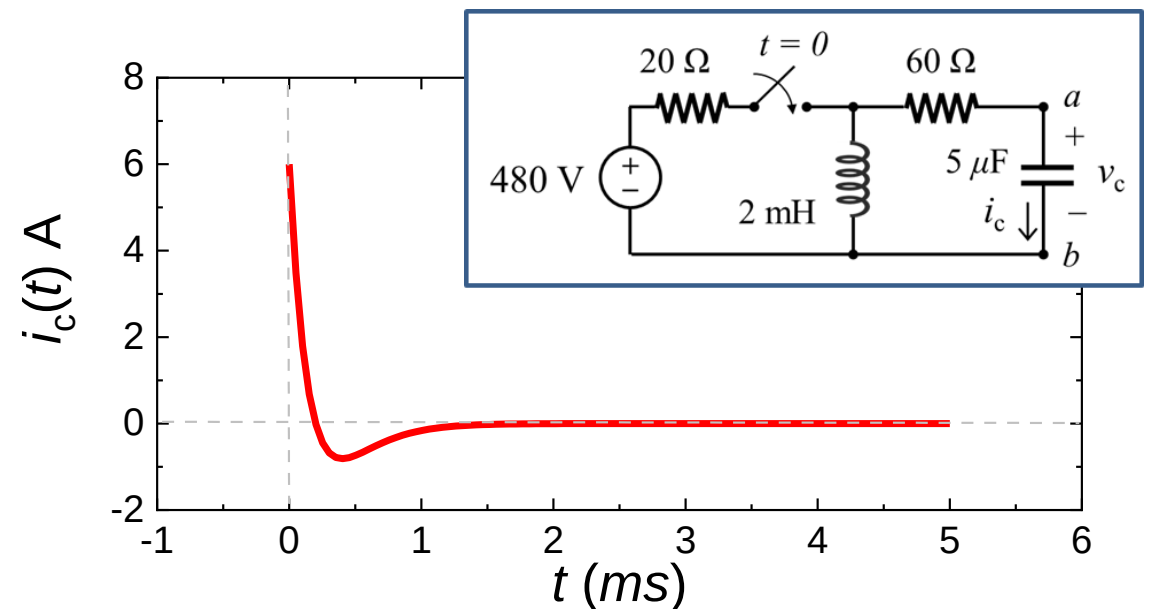
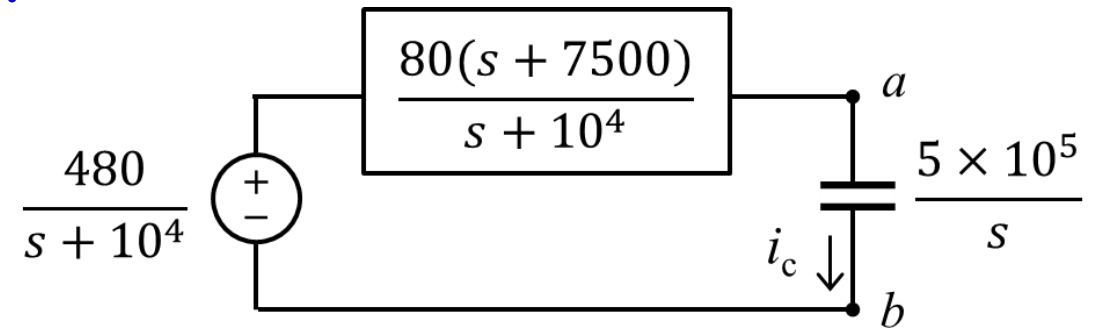
- Example 14.04 (Thevenin theorem):

Example 14.04: Find $i_c(t)$ in the following circuit, assuming the energy stored in the circuit prior to the closing is zero.

Solution: (3) solve for $i_c(t)$:

$$I_C = \frac{6s}{(s + 5000)^2} = \frac{-30000}{(s + 5000)^2} + \frac{6}{s + 5000}$$

$$i_c(t) = (-30000e^{-5000t} + 6e^{-5000t})u(t)$$

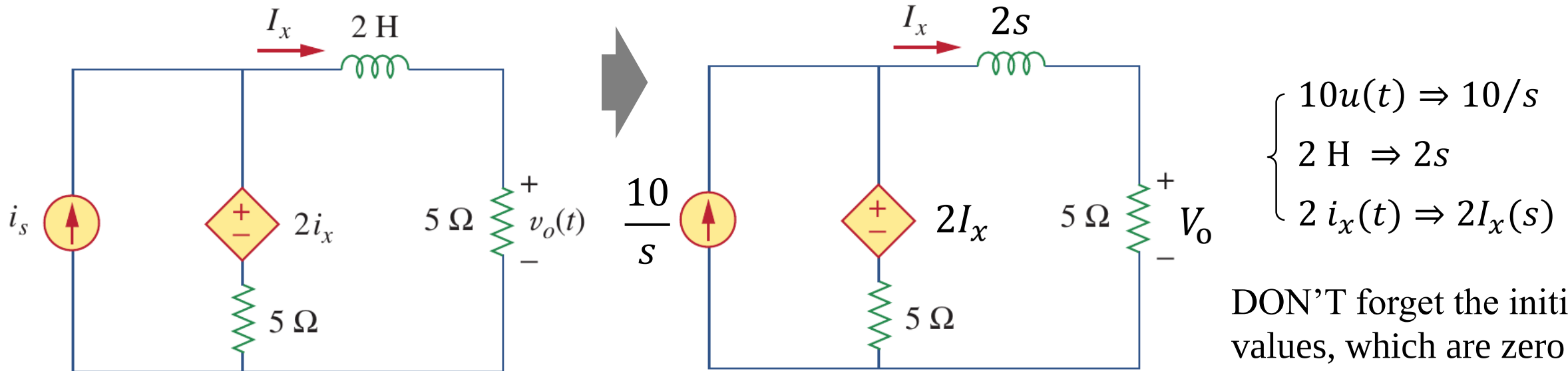


14.3 Circuit theorems in s domain (j)

- Example 14.05 (Thevenin theorem):

Example 14.05: Assume that there is no initial energy stored in the following circuit at $t = 0$ and that $i_s = 10u(t)$ A. (a) Find $V_o(s)$ using Thevenin's theorem. (b) Apply the initial- and final-value theorems to find $v_o(0)$ and $v_o(\infty)$. (c) Determine $v_o(t)$.

Solution: (1) find the equivalent in the s domain:

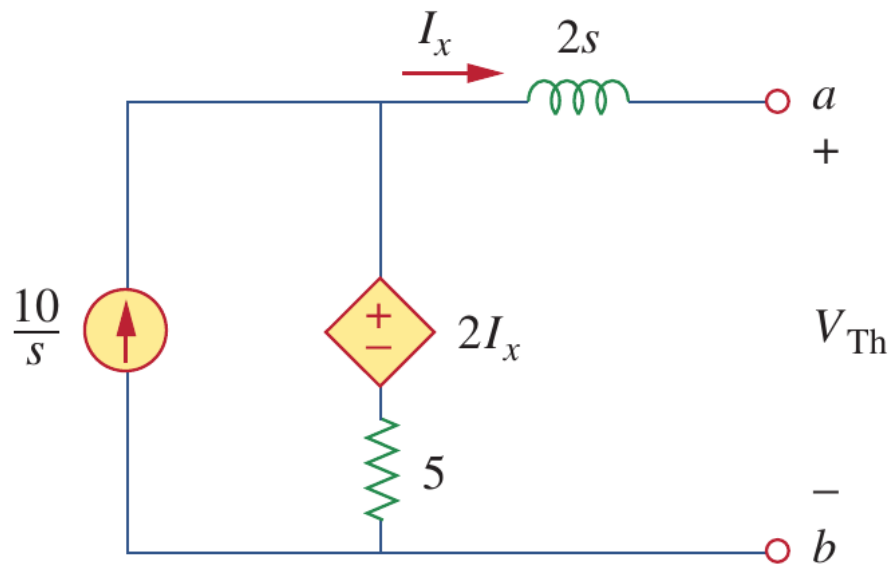


DON'T forget the initial values, which are zero here.

14.3 Circuit theorems in s domain (j)

- Example 14.05 (Thevenin theorem):

Solution: (2) use Thevenin theorem to solve for $V_o(s)$: to get V_{Th}



$$I_x = 0 \Rightarrow V_{oc} = V_{Th} = 5 \left(\frac{10}{s} \right) = \frac{50}{s}$$

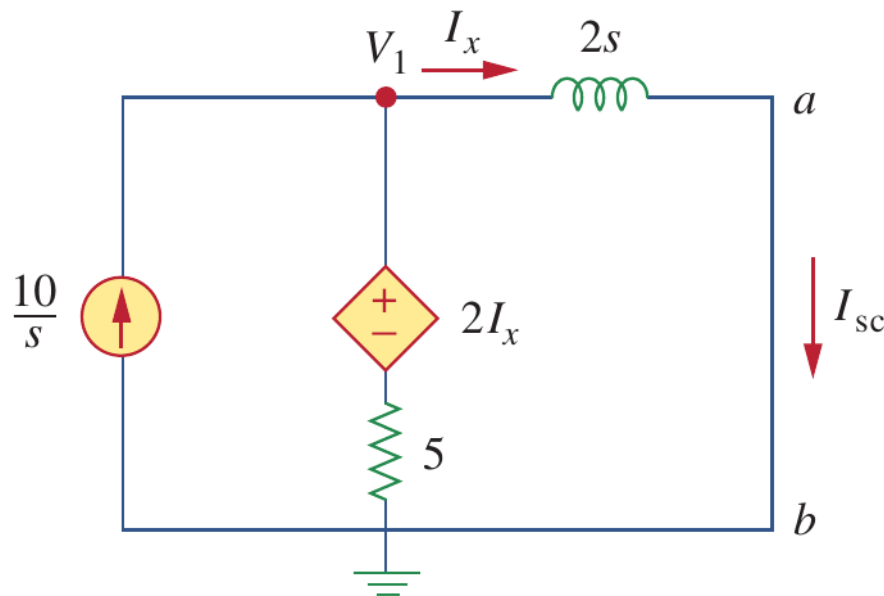
- The next step is usually to get the Thevenin impedance by switching off the independent sources and add an voltage or current source at terminals a and b .
- Another way is the get short-circuit current and then you have $Z_{Th} = V_{oc}/I_{sc}$

14.3 Circuit theorems in s domain (k)

- Example 14.05 (Thevenin theorem):

Solution: (3) use Thevenin theorem to solve for $V_o(s)$: to get Z_{Th}

Using nodal analysis:



$$\begin{cases} -\frac{10}{s} + \frac{(V_1 - 2I_x) - 0}{5} + \frac{V_1 - 0}{2s} = 0 \\ I_x = \frac{V_1}{2s} \end{cases}$$

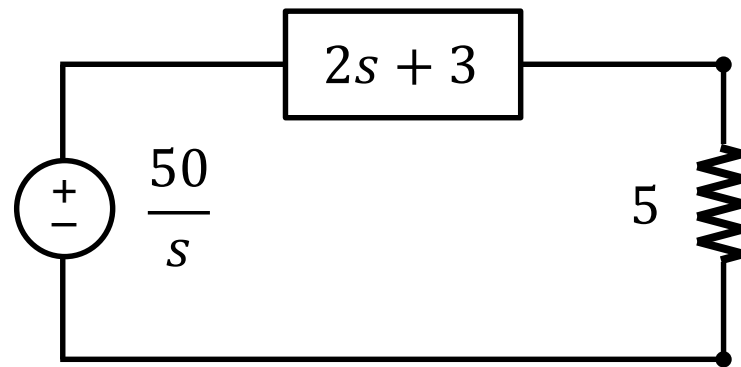
$$\Rightarrow I_{sc} = \frac{V_1}{2s} = \frac{100/(2s + 3)}{2s} = \frac{50}{s(2s + 3)}$$

$$\Rightarrow Z_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{50/s}{50/[s(2s + 3)]} = 2s + 3$$

14.3 Circuit theorems in s domain (I)

- **Example 14.05 (Thevenin theorem):**

Solution: (2) use Thevenin theorem to solve for $V_o(s)$ (**Question (a)**):



$$V_o = \frac{5}{5 + Z_{Th}} V_{Th} = \frac{5}{5 + 2s + 3} \left(\frac{50}{s} \right) = \frac{250}{s(2s + 8)} = \frac{125}{s(s + 4)}$$

(3) Find initial and final values (**Question (b)**):

$$\begin{cases} v_o(0) = \lim_{s \rightarrow \infty} sV_o(s) = \lim_{s \rightarrow \infty} \frac{125}{s + 4} = \lim_{s \rightarrow \infty} \frac{125/s}{1 + 4/s} = \frac{0}{1} = 0 \\ v_o(\infty) = \lim_{s \rightarrow 0} sV_o(s) = \lim_{s \rightarrow 0} \frac{125}{s + 4} = \frac{125}{4} = 31.25 \text{ V} \end{cases}$$

Check the poles before
applying the final
value theorem

14.3 Circuit theorems in s domain (m)

- **Example 14.05 (Thevenin theorem):**

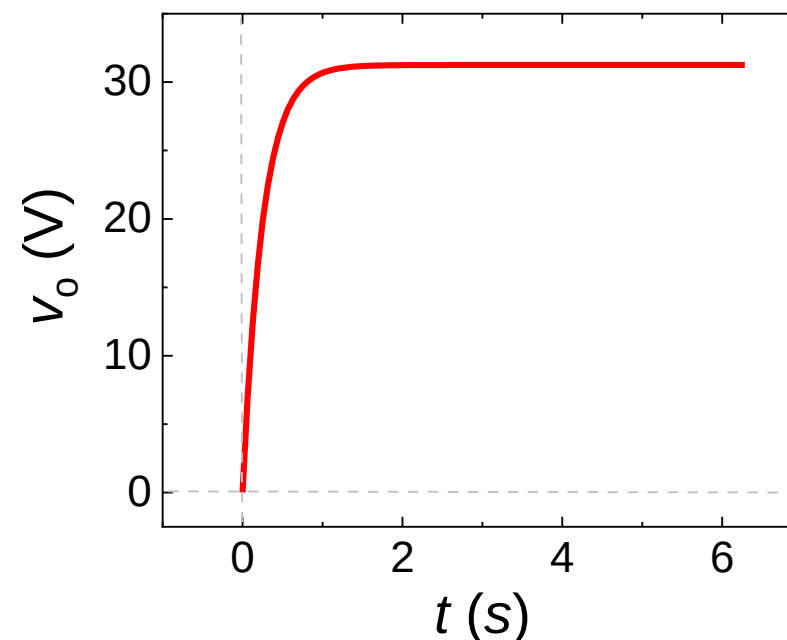
Solution: (4) calculate $v_o(t)$ using inverse Laplace transform (**Question (c)**):

$$V_o = \frac{125}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

$$\begin{cases} A = sV_o(s) \Big|_{s=0} = \frac{125}{s+4} \Big|_{s=0} = 31.25 \\ B = (s+4)V_o(s) \Big|_{s=-4} = \frac{125}{s} \Big|_{s=-4} = -31.25 \end{cases}$$

$$\Rightarrow V_o = \frac{31.25}{s} - \frac{31.25}{s+4}$$

$$\Rightarrow v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$$



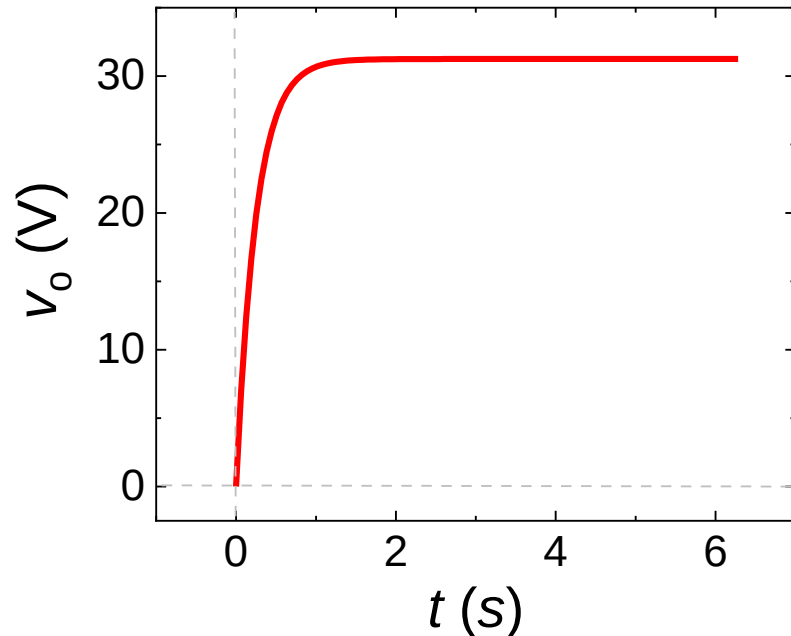
14.3 Circuit theorems in s domain (n)

- **Example 14.05 (Thevenin theorem):**

Solution: (5) Check your results using initial- and final-value theorem:

$$v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$$

The initial- and final-value theorem provide a way to check you calculation.



	From $v_o(t)$	From initial- and final-value theorems	check
$v(0)$	0	0	✓
$v(\infty)$	31.25	31.25	✓

Lesson 14 Applications of Laplace transform

14.1	Elements and laws in s domain	Slides 08-16
14.2-14.3	Circuit analysis in s domain	Slides 18-46
14.4	Transfer functions in s domain	Slides 48-58
14.5	Transformers in s domain	Slides 60-65
14.6	More applications	Slides 67-73
14.7	Summary and assignment	Slides 75-76

14.4 Transfer functions in s domain (a)

- Transfer functions in s domain:

- a) The transfer function is a key concept in signal processing, describing how the output behaves with respect to the input.
- b) The transfer function $H(s)$ is the ratio of the output response $Y(s)$ to the input excitation $X(s)$, assuming all initial conditions are zero.
- c) There are two ways find the transfer function of a circuit: assume input to determine output; ladder method – assume output then walk backwards to find input.

$$H(s) = \frac{Y(s)}{X(s)}$$

$$H(s) = \text{Voltage gain} = \frac{V_o(s)}{V_i(s)}$$

$$H(s) = \text{Current gain} = \frac{I_o(s)}{I_i(s)}$$

$$H(s) = \text{Impedance} = \frac{V(s)}{I(s)}$$

$$H(s) = \text{Admittance} = \frac{I(s)}{V(s)}$$

14.4 Transfer functions in s domain (b)

- The impulse response:

$$H(s) = \frac{Y(s)}{X(s)} \Rightarrow Y(s) = H(s)X(s)$$

When the input is the unit impulse function, $x(t) = \delta(t)$, so that $X(s) = 1$:

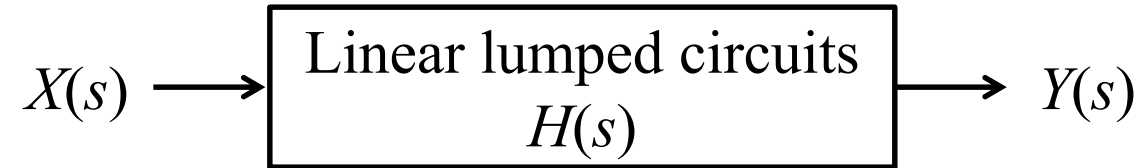
$$\begin{cases} y(t) = h(t), \text{ impulse response} \\ y(t) = \mathcal{L}^{-1}(Y(s)) = \mathcal{L}^{-1}(H(s)) \end{cases}$$

$$\Rightarrow \mathcal{L}^{-1}(H(s)) = h(t)$$

- a) The term $h(t)$ represents the impulse response — it is the time-domain response of the network to a unit impulse.
- b) $H(s)$ is the Laplace transform of $h(t)$ of the network.
- c) Once we know the impulse response $h(t)$ of a network, we can obtain the response of the network to any input signal in the s -domain using this equation or using the convolution integral in the time domain.

14.4 Transfer functions in s domain (c)

- Poles and responses:



Expand $Y(s)$ and assume that there are four terms, in which poles $s = a$ and $s = b$ are from $X(s)$, while poles $s = c$ and $s = d$ are from $H(s)$

$$Y(s) = X(s)H(s) = \frac{A}{s-a} + \frac{B}{s-b} + \frac{C}{s-c} + \frac{D}{s-d}$$

After the inverse
Laplace transform:

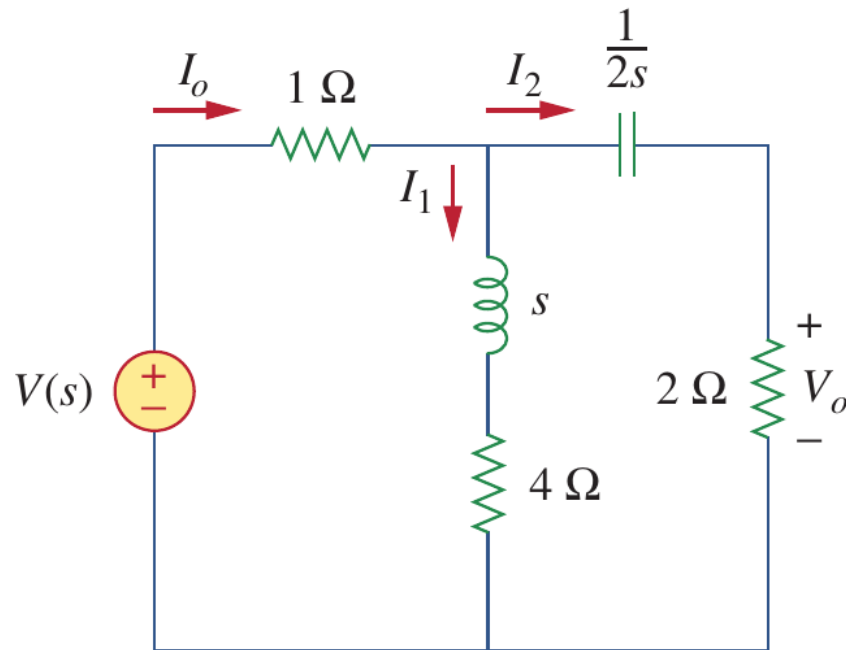
The steady-state
response risen from
the poles of $X(s)$

The transient
response risen from
the poles of $H(s)$

14.4 Transfer functions in s domain (d)

- Example 14.06:

Example 14.06: Determine the transfer function $H(s) = V_o(s)/I_o(s)$ of the following circuit.



Solution I: from the input to the output

$$I_2 = \frac{(s + 4)I_o}{s + 4 + 2 + 1/2s} \quad (\text{current division})$$

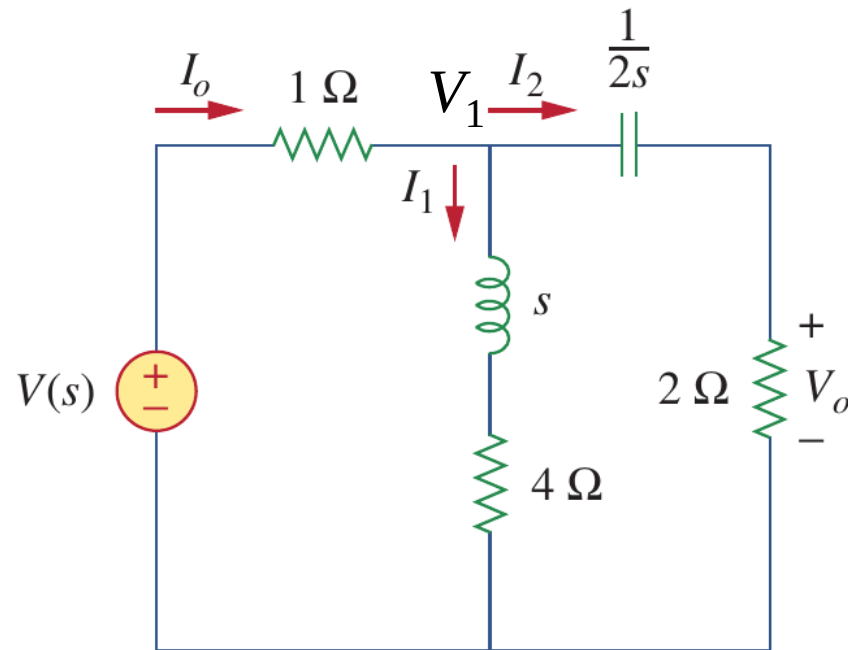
$$V_o = 2I_2 = \frac{2(s + 4)I_o}{s + 6 + 1/2s} \quad (\text{Ohm's Law})$$

$$H(s) = \frac{V_o(s)}{I_o(s)} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

14.4 Transfer functions in s domain (e)

- Example 14.06:

Example 14.06: Determine the transfer function $H(s) = V_o(s)/I_o(s)$ of the following circuit.



Solution II: from the output to the input

Let $V_o = 1$ V, then $I_o = 0.5$ A

$$V_1 = I_2 \left(2 + \frac{1}{2s} \right) = \frac{4s + 1}{4s} \Rightarrow I_1 = \frac{V_1}{s + 4} = \frac{4s + 1}{4s(s + 4)}$$

$$\Rightarrow I_o = I_1 + I_2 = \frac{4s + 1}{4s(s + 4)} + \frac{1}{2} = \frac{2s^2 + 12s + 1}{4s(s + 4)}$$

$$\Rightarrow H(s) = \frac{V_o}{I_o} = \frac{1}{I_o} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

14.4 Transfer functions in s domain (f)

- Example 14.07:

Example 14.07: The output of a linear system is $y(t) = 10e^{-t}\cos(4t)u(t)$ when the input is $x(t) = e^{-t}u(t)$. Find the transfer function of the system in s domain and its impulse response.

Solution:

If $x(t) = e^{-t}u(t)$ and $y(t) = 10e^{-t}\cos 4t u(t)$, then

$$X(s) = \frac{1}{s+1} \quad \text{and} \quad Y(s) = \frac{10(s+1)}{(s+1)^2 + 4^2}$$

Hence,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{10(s+1)^2}{(s+1)^2 + 16} = \frac{10(s^2 + 2s + 1)}{s^2 + 2s + 17}$$

To find $h(t)$, we write $H(s)$ as

$$H(s) = 10 - 40 \frac{4}{(s+1)^2 + 4^2}$$

From Table 15.2, we obtain

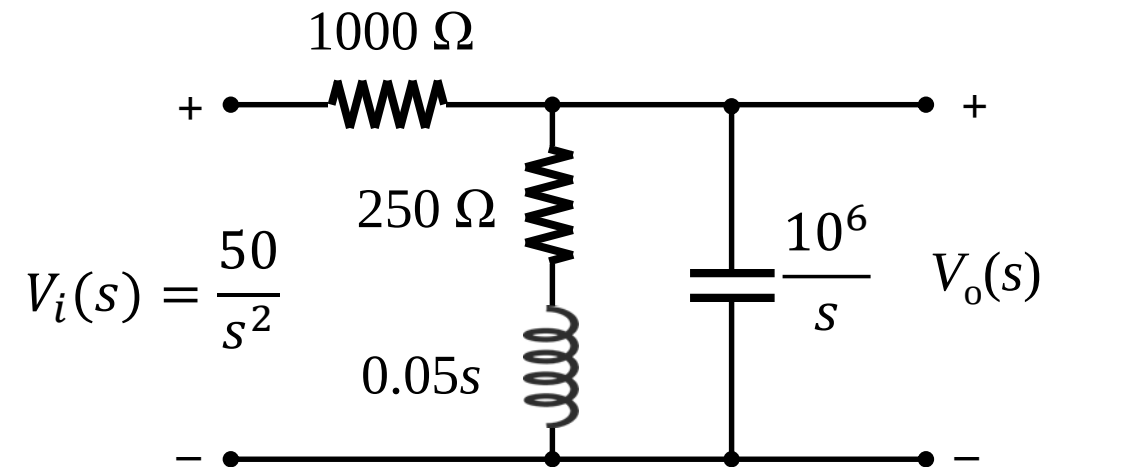
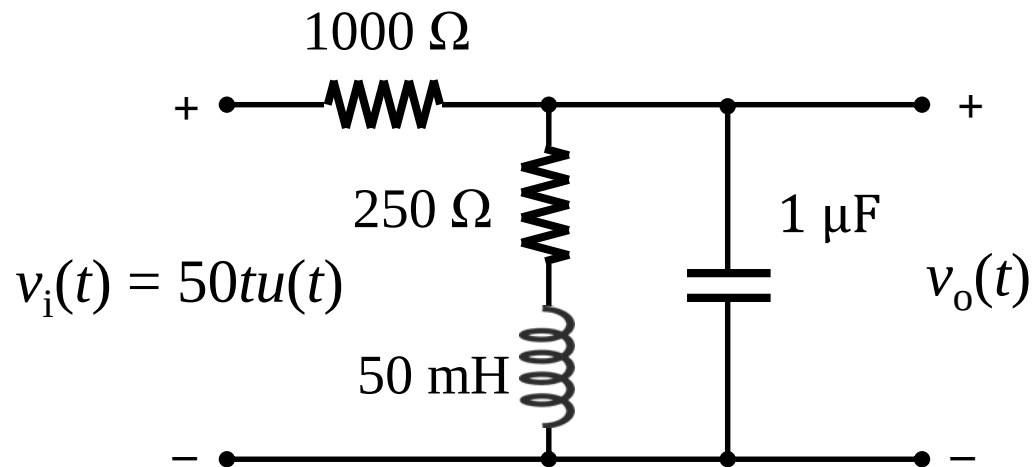
$$h(t) = 10\delta(t) - 40e^{-t} \sin 4t u(t)$$

14.4 Transfer functions in s domain (g)

- Example 14.08:

Example 14.08: The input voltage is $v_i(t) = 50tu(t)$. Find (a) $v_o(t)$ using the transfer function, (b) the transient response $v_t(t)$ and (c) the steady-state response $v_{ss}(t)$. Assuming zero initial conditions.

Solution: (1) find the equivalent in the s domain:



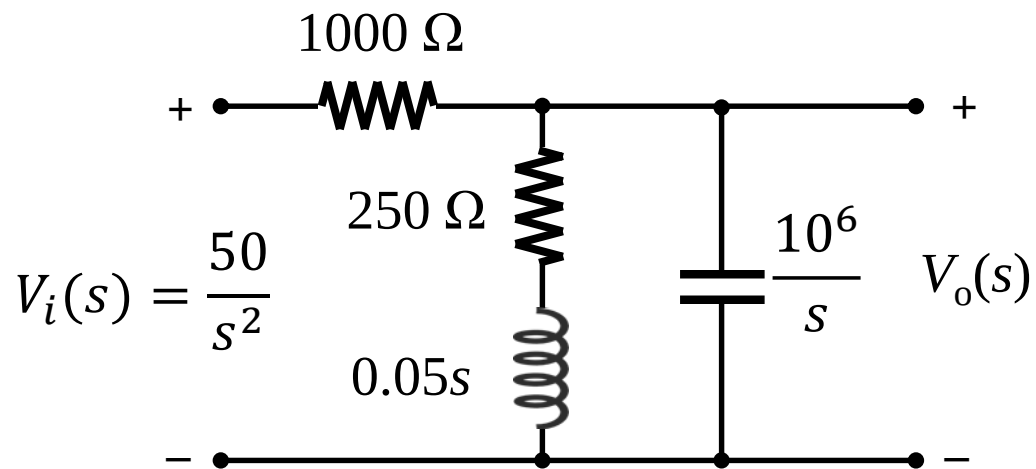
$$50tu(t) \Rightarrow \frac{50}{t^2} \quad 50 \text{ mH} \Rightarrow 0.05s \quad 1 \mu\text{F} \Rightarrow \frac{10^6}{s}$$

14.4 Transfer functions in s domain (h)

- Example 14.08:

Example 14.08: The input voltage is $v_i(t) = 50tu(t)$. Find (a) $v_o(t)$ using the transfer function, (b) the transient response $v_t(t)$ and (c) the steady-state response $v_{ss}(t)$. Assuming zero initial conditions.

Solution: (2) find transfer function in the s domain using nodal analysis



$$\frac{V_i - V_o}{1000} - \frac{V_o}{250 + 0.05s} - \frac{sV_o}{10^6} = 0$$

$$\Rightarrow V_o(s) = \frac{1000(s + 5000)V_i}{s^2 + 6000s + 26 \times 10^6}$$

$$\Rightarrow H(s) = \frac{V_o}{V_i} = \frac{1000(s + 5000)}{s^2 + 6000s + 26 \times 10^6}$$

$$\text{Therefore, } H(s) = \frac{1000(s + 5000)}{(s + 3000 - j4000)(s + 3000 + j4000)}$$

14.4 Transfer functions in s domain (i)

- **Example 14.08:**

Solution: (3) find $V_o(s)$ and $v_o(t)$

We know that
$$H(s) = \frac{1000(s + 5000)}{(s + 3000 - j4000)(s + 3000 + j4000)}$$

→
$$V_o(s) = H(s)V_i(s) = \frac{1000(s + 5000)}{(s + 3000 - j4000)(s + 3000 + j4000)} \frac{50}{s^2}$$

Partial fraction expansion

$$V_o(s) = \frac{k_1}{(s + 3000 - j4000)} + \frac{k_1^*}{(s + 3000 + j4000)} + \frac{k_2}{s^2} + \frac{k_3}{s}$$

Residue method

$$V_o(s) = \frac{5\sqrt{5} \times 10^{-4} \angle(79.7^\circ)}{(s + 3000 - j4000)} + \frac{5\sqrt{5} \times 10^{-4} \angle(-79.7^\circ)}{(s + 3000 + j4000)} + \frac{10}{s^2} + \frac{-4 \times 10^{-4}}{s}$$

Inverse LT

$$v_o(t) = [10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.7^\circ) + 10t - 4 \times 10^{-4}] u(t) \text{ V}$$

14.4 Transfer functions in s domain (j)

- Example 14.08:

Solution: (4) find $v_t(t)$ and $v_{ss}(t)$

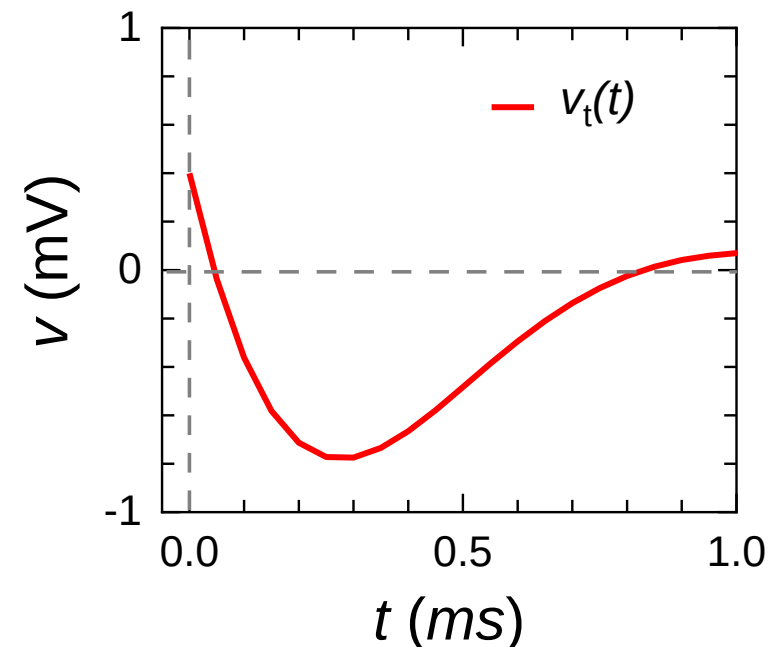
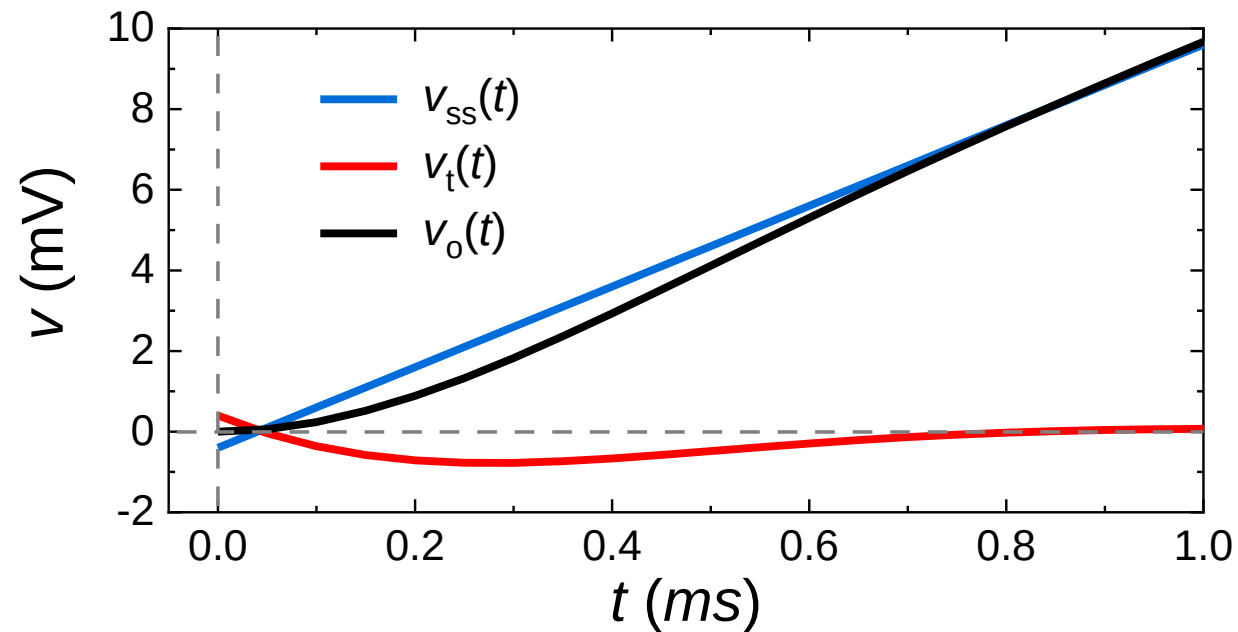
$$V_o(s) = H(s)V_i(s) = \frac{1000(s + 5000)}{(s + 3000 - j4000)(s + 3000 + j4000)} \frac{50}{s^2}$$
$$V_o(s) = \frac{5\sqrt{5} \times 10^{-4} \angle(79.7^\circ)}{(s + 3000 - j4000)} + \frac{5\sqrt{5} \times 10^{-4} \angle(-79.7^\circ)}{(s + 3000 + j4000)} + \frac{10}{s^2} + \frac{-4 \times 10^{-4}}{s}$$
$$v_o(t) = \underbrace{[10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.7^\circ)]}_{\text{Transient response}} + \underbrace{[10t - 4 \times 10^{-4}]}_{\text{SS response}} u(t) \text{ V}$$

$$\begin{cases} v_t(t) = [10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.7^\circ)] u(t) \text{ V, rising from poles of } H(s) \\ v_{ss}(t) = [10t - 4 \times 10^{-4}] u(t) \text{ V, rising from poles of the input signal} \end{cases}$$

14.4 Transfer functions in s domain (k)

- Example 14.08:

Solution: (5) Here you can see the wave forms



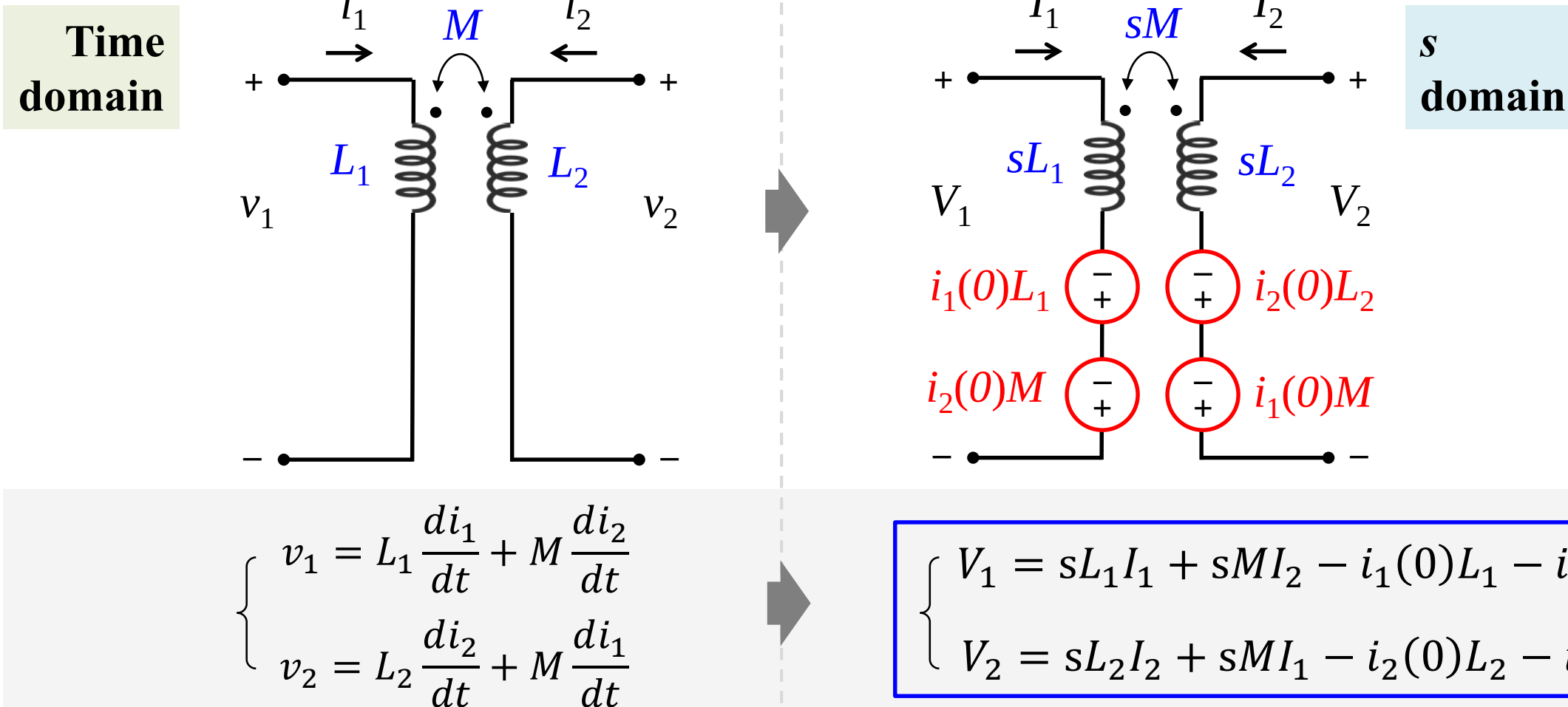
→
$$\begin{cases} v_t(t) = [10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.7^\circ)]u(t) \text{ V, rising from poles of } H(s) \\ v_{ss}(t) = [10t - 4 \times 10^{-4}]u(t) \text{ V, rising from poles of the input signal} \end{cases}$$

Lesson 14 Applications of Laplace transform

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14.5 Transformers in s domain (a)

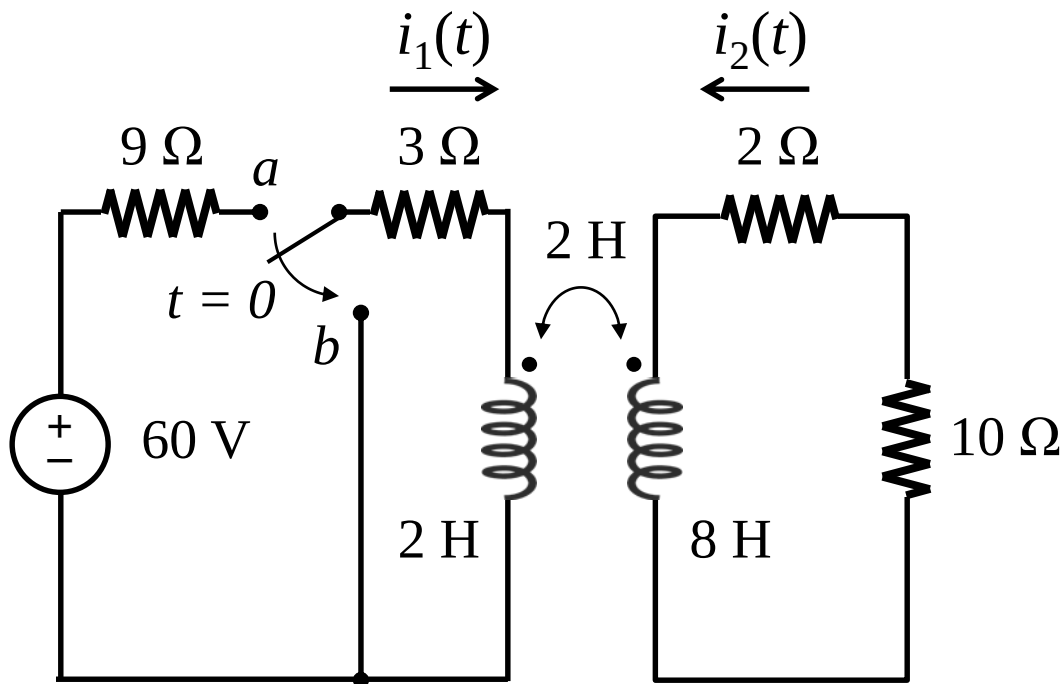
- Transformers in s domain:



14.5 Transformers in s domain (b)

- Example 14.09:

Example 14.09: In the circuit shown below, the switch has been at position *a* for a long time. At $t = 0$, switch is moved to position *b*. find $i_2(t)$.



Solution: (1) find initial values

Before the step at $t = 0$:

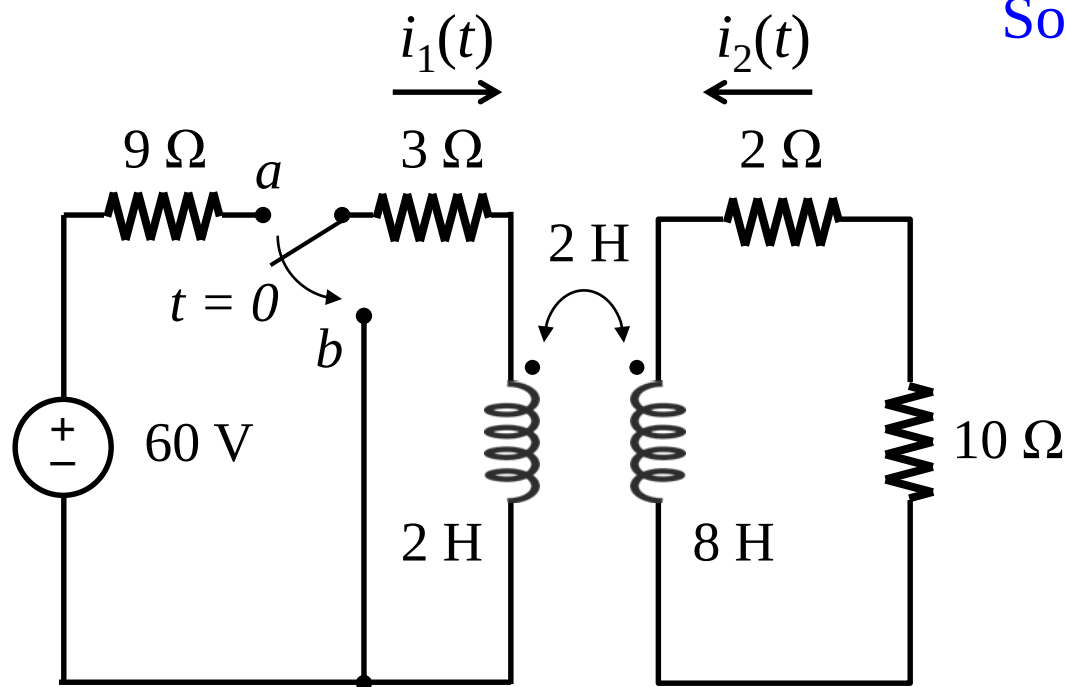
$$\begin{cases} i_1(0) = 60/12 = 5 \text{ A} \\ i_2(0) = 0 \text{ A} \end{cases}$$

If the question doesn't tell you the initial values, then you need to do steady-state analysis to find them, just like we did in Lessons 7 and 8.

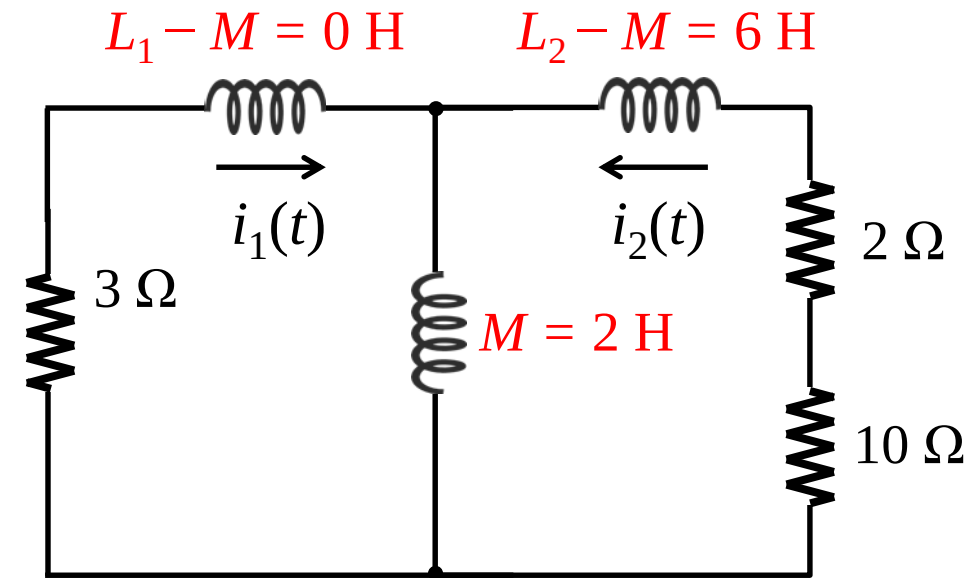
14.5 Transformers in s domain (c)

- Example 14.09:

Example 14.09: In the circuit shown below, the switch has been at position *a* for a long time. At $t = 0$, switch is moved to position *b*. find $i_2(t)$.



Solution: (2) find the decoupled equivalent circuit at $t = 0$:

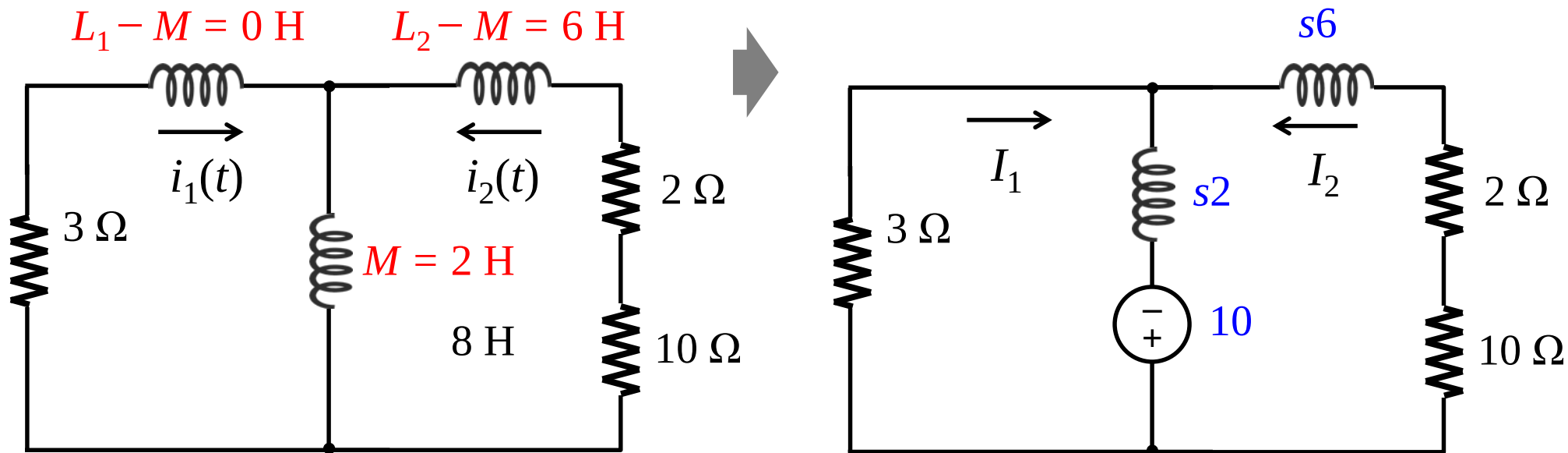


14.5 Transformers in s domain (d)

- Example 14.09:

Example 14.09: In the circuit shown below, the switch has been at position *a* for a long time. At $t = 0$, switch is moved to position *b*. find $i_2(t)$.

Solution: (3) find the equivalent circuit in the s domain:



$$\begin{cases} i_1(0) = 5 \text{ A} \\ i_2(0) = 0 \text{ A} \end{cases}$$

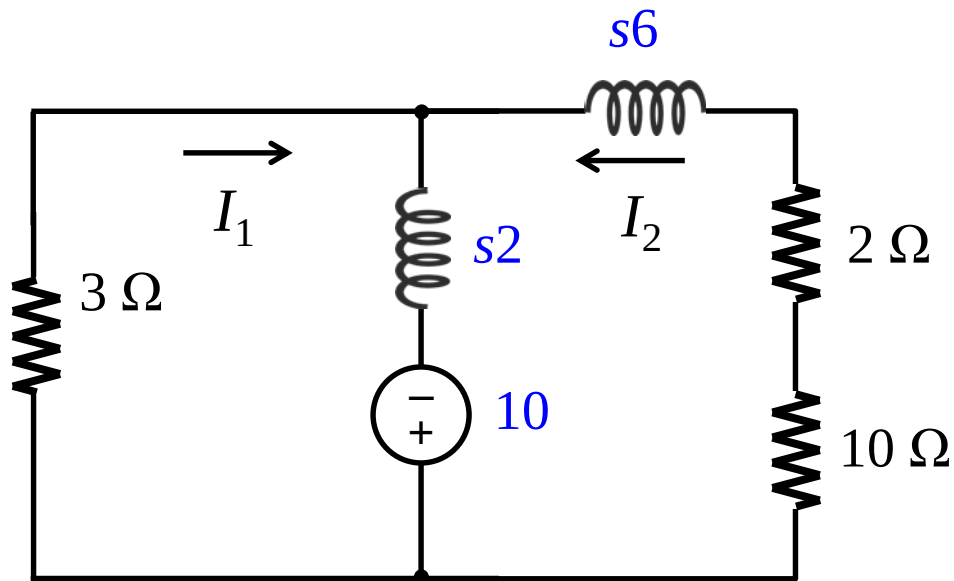
Why three inductors but only one V source?

14.5 Transformers in s domain (e)

- Example 14.09:

Example 14.09: In the circuit shown below, the switch has been at position *a* for a long time. At $t = 0$, switch is moved to position *b*. find $i_2(t)$.

Solution: (4) Solve for I_2 and $i_2(t)$:



Using mesh analysis:

Why positive?

$$\begin{cases} (3 + 2s)I_1 + 2sI_2 = 10 \\ +2sI_1 + (12 + 8s)I_2 = 10 \end{cases}$$

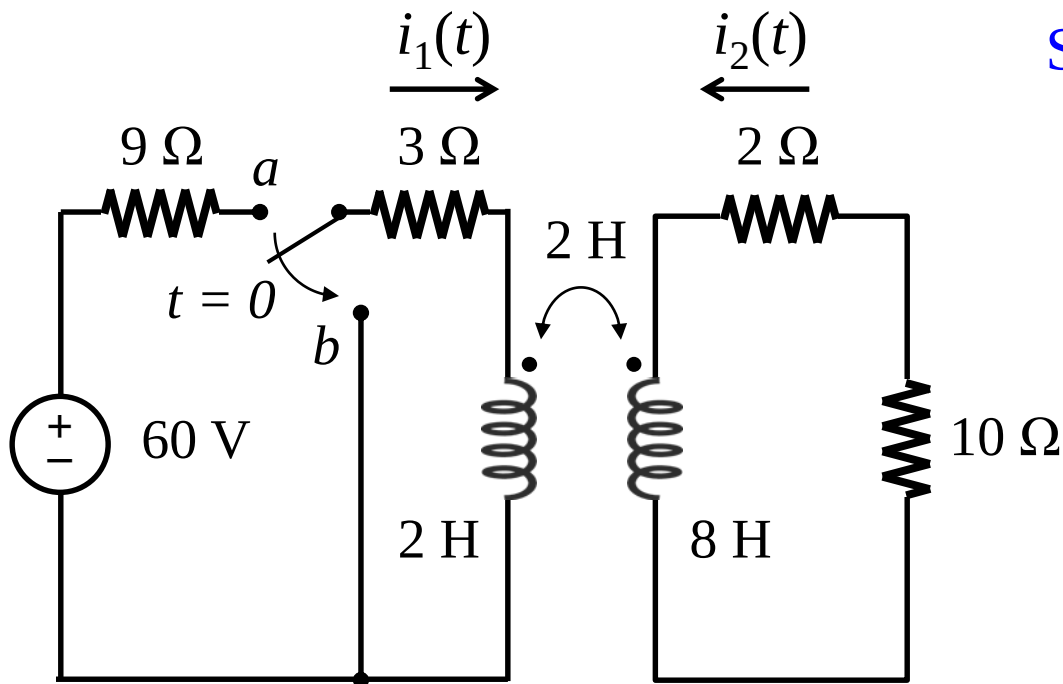
$$I_2 = \frac{2.5}{(s + 1)(s + 3)} = \frac{1.25}{s + 1} - \frac{1.25}{s + 3}$$

$$i_2(t) = (1.25e^{-t} - 1.25e^{-3t})u(t) \text{ A}$$

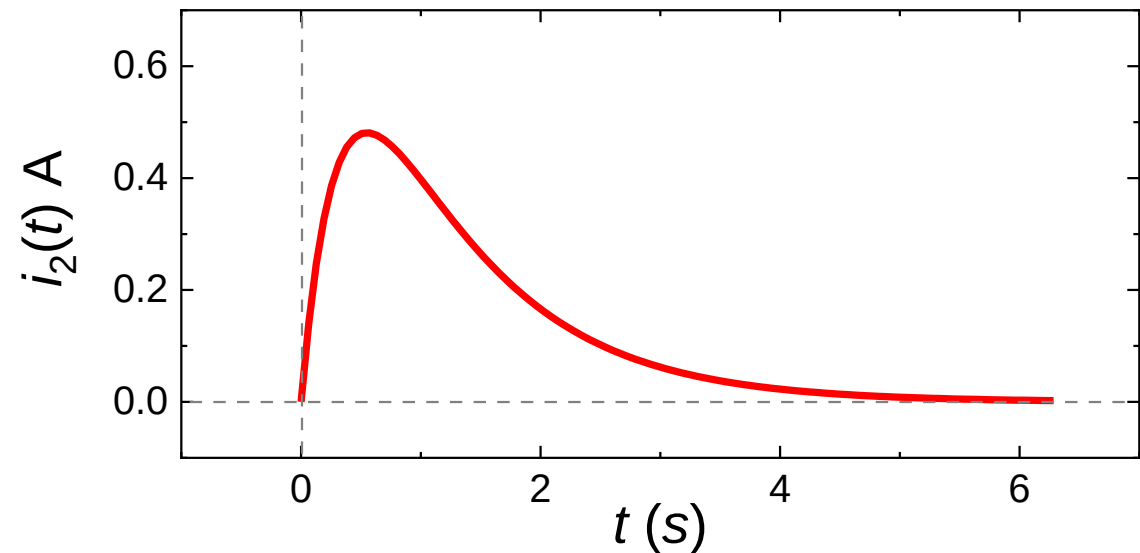
14.5 Transformers in s domain (f)

- Example 14.09:

Example 14.09: In the circuit shown below, the switch has been at position *a* for a long time. At $t = 0$, switch is moved to position *b*. find $i_2(t)$.



Solution: (5) Here you can see the waveform:



Lesson 14 Applications of Laplace transform

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14.6 System stability (a)

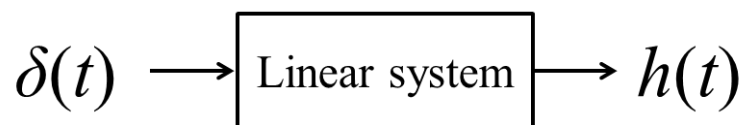
- Recall: advantages of Laplace transform in circuit analysis

The Laplace transform is of vast importance:

- ☒ It can be applied to a wider variety of inputs (non-sinusoidal) than phasor analysis.
- ☒ It provides an easy way to solve circuit problems involving initial conditions.
- ☒ The total response can be obtained in a single step.
- ☐ Transfer function defined in s -domain reflects the circuit properties such as stability.

14.6 System stability (b)

- System stability:



$$\lim_{t \rightarrow \infty} |h(t)| = \text{finite}$$

Stability of a circuit/system:

- a) A circuit is **stable** if its impulse response $h(t)$ is bounded (有界的, i.e., $h(t)$ converges to a finite value) as $t \rightarrow \infty$.
- b) In contrast, the circuit is **unstable** if $h(t)$ grows without bound (无界的) as $t \rightarrow \infty$.

$$\mathcal{L}(h(t)) = H(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}$$

This means the Laplace transform the impulse response, namely the transfer function, must satisfy following conditions:

$m > n$ & poles are in the left half of the s plane.

14.6 System stability (c)

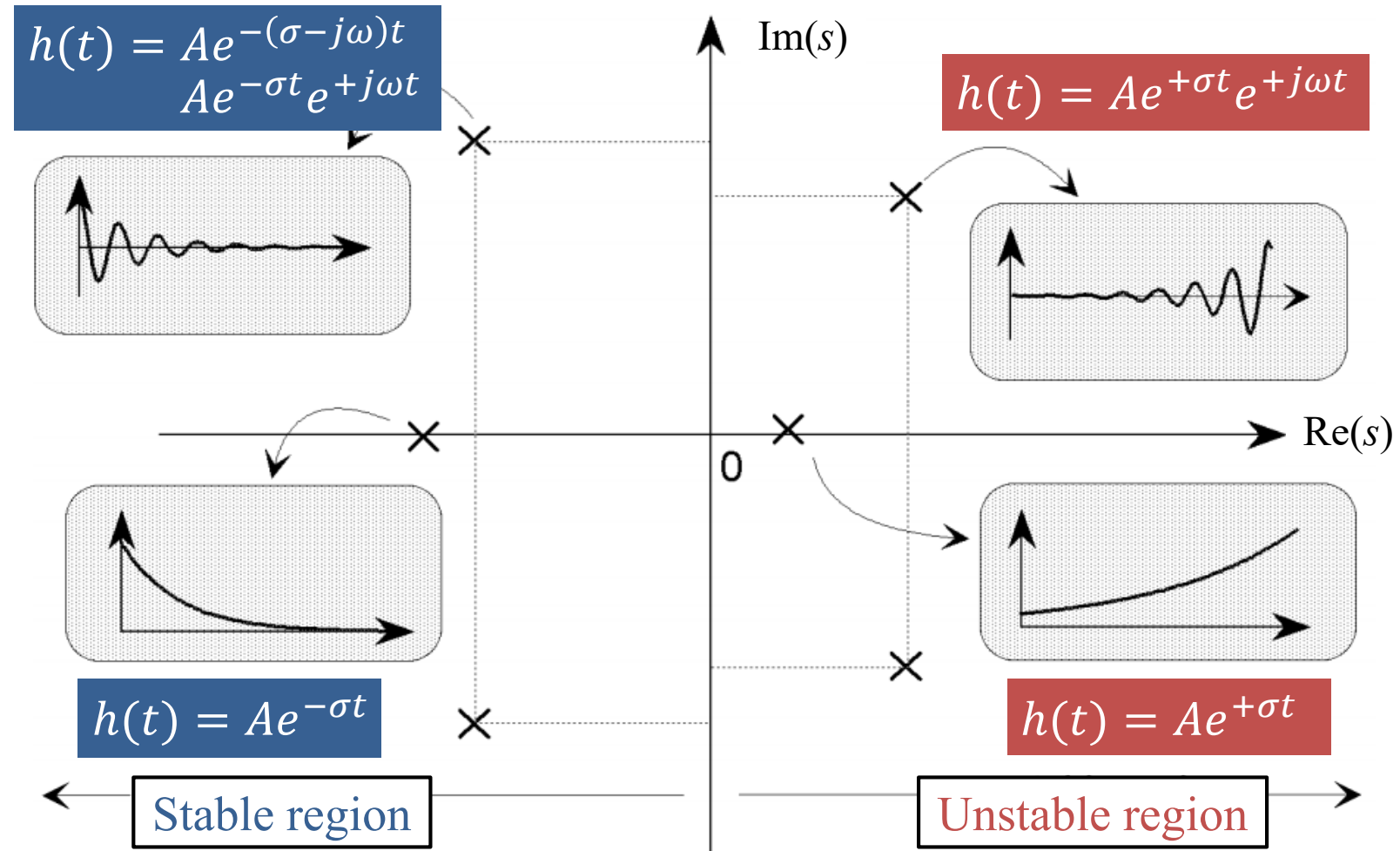
- Why “poles are in the left half of the s plane”?

Assuming a certain transfer function can be expanded as shown below:

$$H(s) = \frac{N(s)}{D(s)} = \frac{A}{s - p_1},$$

in which $s = \sigma + j\omega$.

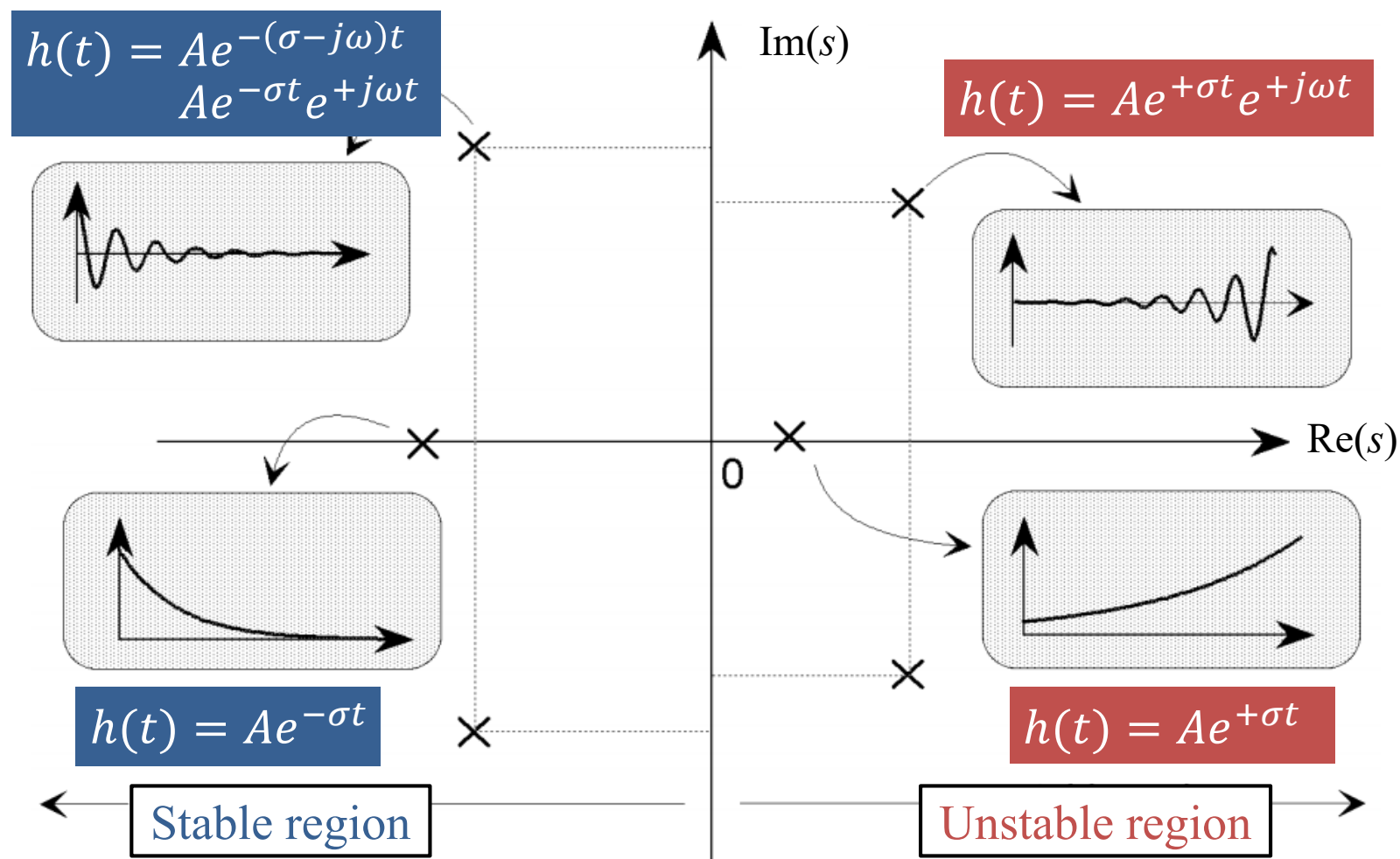
Depending on the location of the poles in the s plane, the inverse Laplace transform of $H(s)$, i.e. the $h(t)$, represents different types of signals



14.6 System stability (d)

- Circuits based on only RLC cannot be unstable:

- a) A circuit made up exclusively of passive elements (R , L , and C) and independent sources cannot be unstable.
- b) Otherwise some branch currents or voltages would grow indefinitely with sources set to zero.
- c) Passive elements cannot generate such indefinite growth. Passive circuits either are stable or have poles with zero real parts.

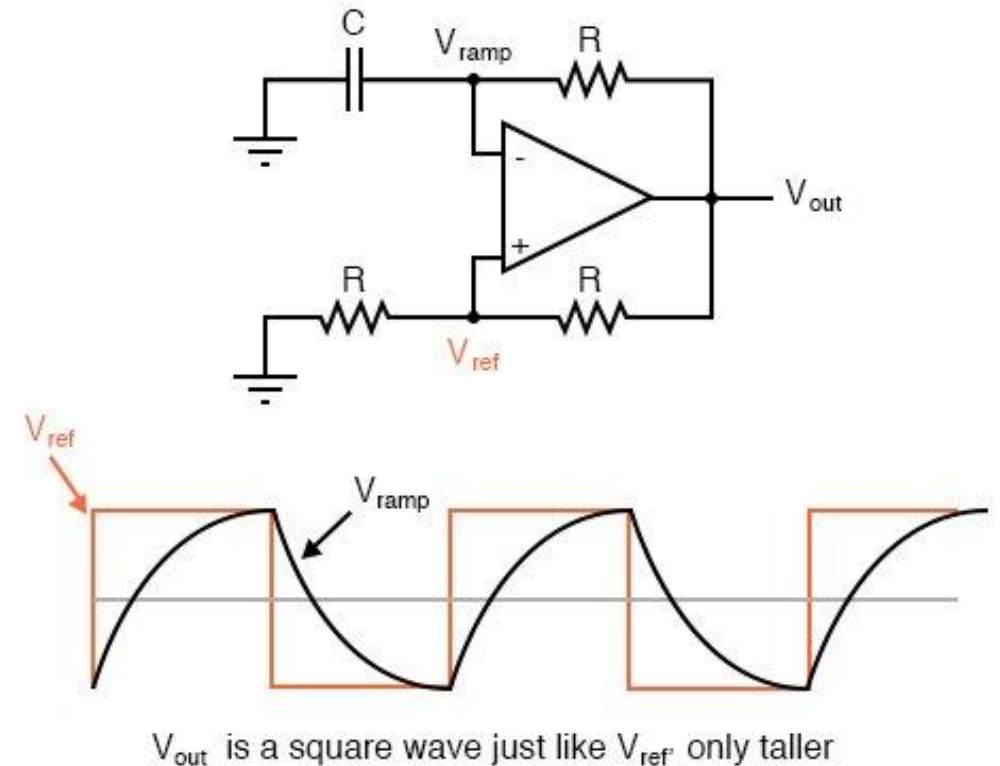


14.6 System stability (e)

- Unstable circuits are also useful:

- a) On the other hand, active circuits or passive circuits with controlled sources can supply energy, therefore they can be unstable.
- b) In fact, an oscillator is a typical example of a circuit designed to be unstable which is very useful.
- c) Comparator based on op-amps with positive feedback is also very useful

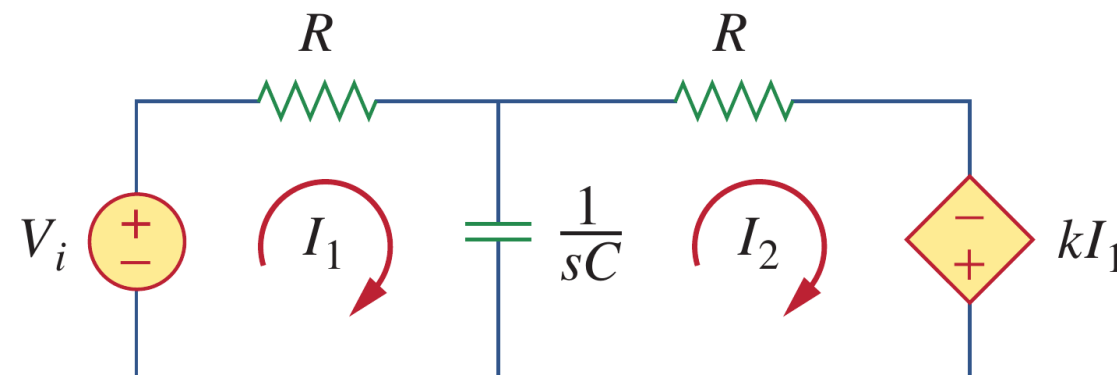
Oscillator circuit using positive feedback



14.6 System stability (f)

- Example 14.10:

Example 14.10: determine the values of k for which the following circuit is stable.



Solution: using mesh analysis

$$\begin{cases} V_i = \left(R + \frac{1}{sC}\right)I_1 - \frac{I_2}{sC} \\ 0 = -kI_1 + \left(R + \frac{1}{sC}\right)I_2 - \frac{I_1}{sC} \end{cases} \Rightarrow \begin{cases} H_1(s) = \frac{I_1}{V_i} = \frac{\frac{sRC + 1}{RC(k + 1)}}{s} \Rightarrow s = 0 \\ H_2(s) = \frac{I_2}{V_i} = \frac{R^2C(skC + 1)}{s + \frac{2R - k}{R^2C}} \Rightarrow s = \frac{k - 2R}{R^2C} \end{cases}$$

When $k < 2R$, stable; when $k > 2R$, unstable

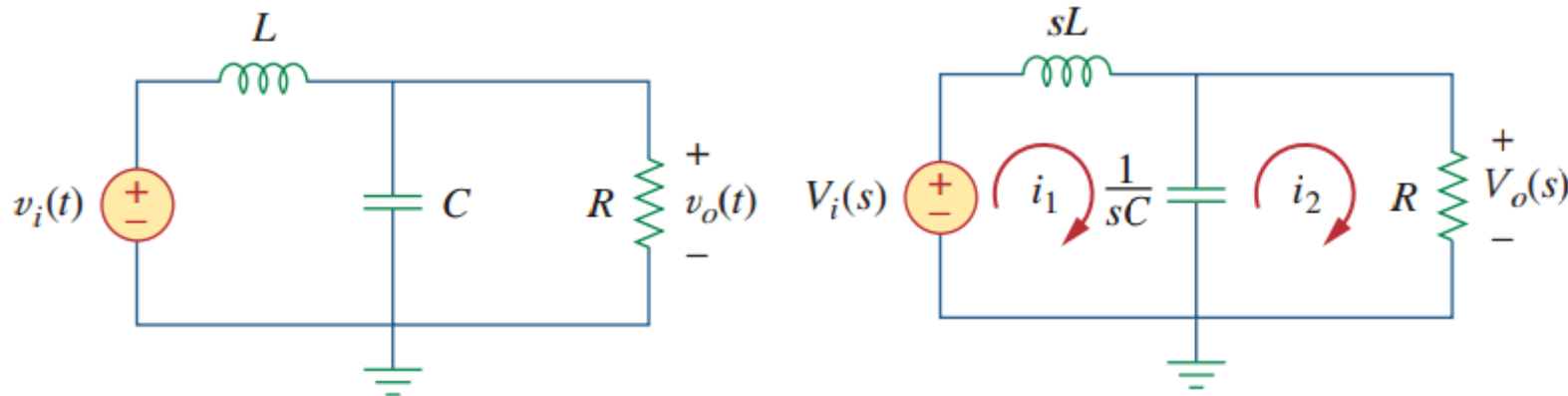
14.6 System synthesis (a)

- System synthesis:

- a) Network synthesis is finding a network that represents a given transfer function.
- b) Text book Pages 740-745.

Given the transfer function $H(s) = \frac{V_o(s)}{V_i(s)} = \frac{10}{s^2 + 3s + 10}$

Implement the function using the circuit below. (a) if $R = 5 \Omega$, find L and C . b) if $R = 1 \Omega$, find L and C .

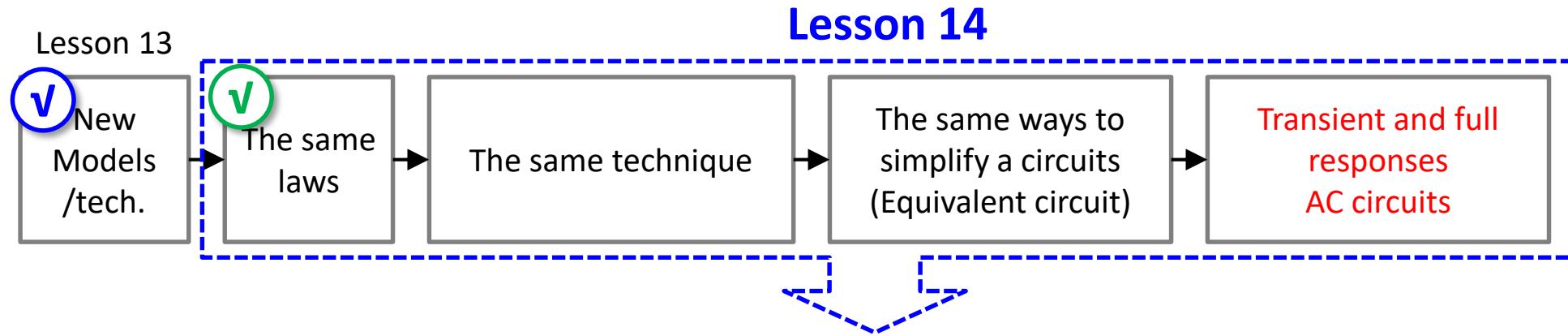


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14.7 summary and assignment (a)

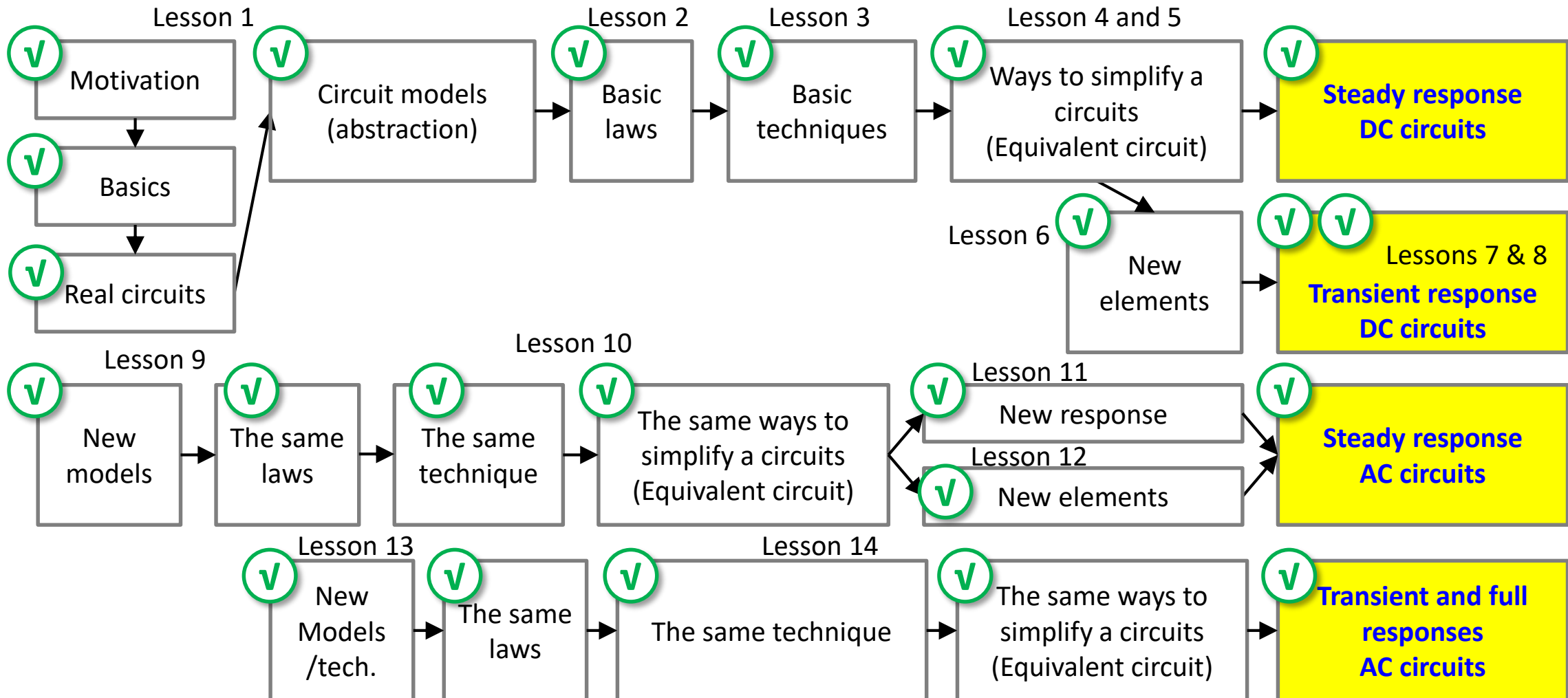
- EE104 road map



1. Elements and laws in s domain.
2. Circuits analysis in s domain
3. Transfer function in s domain
4. Transformers in s domain
5. More applications

14.7 summary and assignment (b)

- EE104 road map



14.7 summary and assignment (c)

These problems are recommended for practice:

16.20, 16.33, 16.57, 16.69, 16.75

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

Arcus clouds

